

CHAPTER ONE

Introduction

Security describes protection of life and property. There are doors to keep people out, Key locks and chains reinforce the mode of security. Doors and windows are being made of metals not just wood anymore. Influential persons in our society have bullet proof doors to ensure a good measure of security of self and family.

The security sector is experiencing diversification as it has never seen before. This has brought about the need to review the reliability of already existing systems and look into the possibility of creating better systems that are smarter and more secure. The micro controller based digital lock presented here is an access control system that allows only authorized persons to access a restricted area (gurusamy, 1920), this system is best suitable for corporate offices, automated machine (ATMs) and home security. It comprises of a small electronic unit which is in fixed at the entry door to control a relay operated lock with the help of a stepper motor, when an authorized person enters predetermined user password via the keypad the stepper motor is operated for a limited time to unlatch the relay-operated lock so the door can be open. At the end of preset delay time, the stepper motor is operated in reverse direction and the door gets locked again.

When the code has been incorrectly entered in a row, the code lock will switch to block mode, this function thwarts any attempt by „hacker to quickly try a large number of codes in a sequence. If the user forgets his password, the code lock can be accessed by a unique digit administrator password and the secret code can be changed any time since we have saved the password in the EEPROM.

The project intends to interface the microcontroller with the LED keypad and other hardware components like EEPROM, sounder start/stop the dc motor through sending a right or wrong password. The measure of efficiency is based on how fast the microcontroller can detect the incoming message and act accordingly.

The Controlling unit has an application program to allow the microcontroller read the incoming data through the modem and control the engine motor as per the requirement. The performance of the design is maintained by the controlling unit. This project uses 8051 microcontroller as the central processing unit. Specifically the proto-type make used of AT89c51 microcontroller with Programs written in C language burnt inside the microcontroller to perform the following capabilities; solid wood door, panel doors, metal skinned wood-edged doors and metal edge wrapped doors.

1.1 Home security system

House security system is one of security that truly related to burglar or safety alarm system. Burglar and safety alarms are found in electronic form nowadays. Sensors are connected to a control unit

via either a low-voltage hardwire which in turn connects to a means for announcing the alarm to elicit response. In a new construction systems are predominately hardwired for economy while in retrofits wireless systems may be more economical and certainly quicker to install. Some systems are dedicated to one mission; handle fire, intrusion, and safety alarms simultaneously.

In common security system, the lights are triggered by motion gives the impression to user that someone is at home and able to see the burglar. Infrared motion detectors placed in house security system in crucial areas of the house can detect any burglars and alert the home owner or police. The first security system invented, house alarms were triggered by the release of a pressure button fitted into a door or window frame. This basic alarm was fundamentally flawed as the entire intruder needed to do to silence. The alarm was to close the door or window. While various systems on the market ranging from inexpensive house security alarms to highly sophisticated systems requiring professional installation. All modern alarms are based on the same foundation, the electric circuit which is completed either when the door is opened or closed depending on the security system designed. The alarm is triggered when the circuit is altered and will not be silenced until a code is punched into the control panel.

The most expensive and complicated alarm systems might also involve a combination of motion sensors and pressure pads to ensure even the most cunning intruder doesn't get his hands on treasures.

1.2 Objective of the project

The objective of the project resembles to the specific objective and general objective. Security is a prime concern in our day-today life. Everyone wants to be as much secure as possible. An access control for doors forms a vital link in a security chain the micro controller based digital lock for Doors is an access control system that allows only authorized persons to access a restricted area. The system is fully controlled by the 8 bit microcontroller AT89C51 which has a 4Kbytes of ROM for the program memory. The password is stored in the EPROM so that we can change it at any time. The system has a keypad by which the password can be entered through it. When they entered password equals with the password stored in the memory then the relay gets on and so that the door is opened. If we entered a wrong password then the pins to the input of the microcontroller get signal automatically the output buzzer that is connected to that input makes sound.

The specific objective of the project is:

1. To minimize man power
2. To minimize the need for a key
3. To keep home security based on microcontroller
4. To use only by the authorized person
5. To know what microcontroller in detail
6. To develop password based door and infrared based window lock security system

1.3 Significance of the project

1. This kind of systems control home security and other systems, in order to improve comfort, energy efficiency.
2. This kind of systems is particularly useful for the disabled or elderly, improving the life quality and avoiding special aid expenses.
3. A Secure home creation
4. Home life span increases
5. Improved Productivity and Cost Avoidance.
6. Improved Security and Service.

1.4 Problems statement

Security is a prime concern in our day-today life. Everyone wants to be as much secure as possible. Knowing your home is protected provides peace of mind both when you are away and when you are home. Security is important even if you have outstanding public safety agencies in your area. There are far more homes than there are police officers, not to mention of skilled thieves. So we would like to implement our project to do everything possible to make your house secure rather than just relying on others.

With the increasing busy schedules people rely on machines to support them in this modern world. It led to the need for intelligence to these machines in every area. Here the case is with home security. Our project involves an access control for doors and windows forming a vital link in a security chain implemented using fully controlled 8 bit microcontrollers 89c51. The Microcontroller based Home Security System can be adopted at Home, it has various types of Sensors. The other microcontroller can be carried along with the user or can be placed at his working place, it has LCD .In our Project we have Infrared sensor (IR sensor).

The Microcontroller at the transmitter end will continuously monitors all the Sensors and if any security problem is found then the Microcontroller at the receiver end will signal to the light emitting diode yellow to indicate there is someone outside and the type of problem is displayed on the LCD. This equipment uses low power and operates in real time.

1.5 Scope of the project

Home security should have double meanings; one refers to the life and safety, two-finger property theft. Traditional home security is to install a passive infrared detector, which is characterized installed in the interior, so the room unattended, its role in the protection of property theft can be achieved. Active infrared detectors, and its strong anti-interference performance, and also mounted on the outside, the indoor human activity has nothing to do, as long as the external entrance of your house equipped with active infrared detectors, once someone enter the room, on the timely warning, life security is guaranteed, the property is intact. Play a defensive back, the consequences of a small purpose.

1.5.1 Applications

- Apartment house
- Banks
- Hospitals
- Industry
- Stores,
- In big governmental offices
- Even in personal homes.

1.6 Material required

- 8051 microcontroller
- LCD (liquid crystal display)
- IR sensor
- Keypad
- Crystal oscillator
- LED
- Buzzer
- Motor drive
- Dc motor
- EEPROM
- Capacitor

1.7 Methodology

Steps that we follow when we are doing this project are listed as follows:

- Searching of materials used for our project through reading books, referring different websites, and by asking our advisor concerned to the provided project title.
- Arrange the data that we have already collected.
- Build a schematic diagram of a system.
- Design the PIC microcontroller program for home security system.
- Simulate the PIC program using software.
- Show the output after simulation (implement the simulation)

Simple systematic diagram which the methodology that we planned to follow.

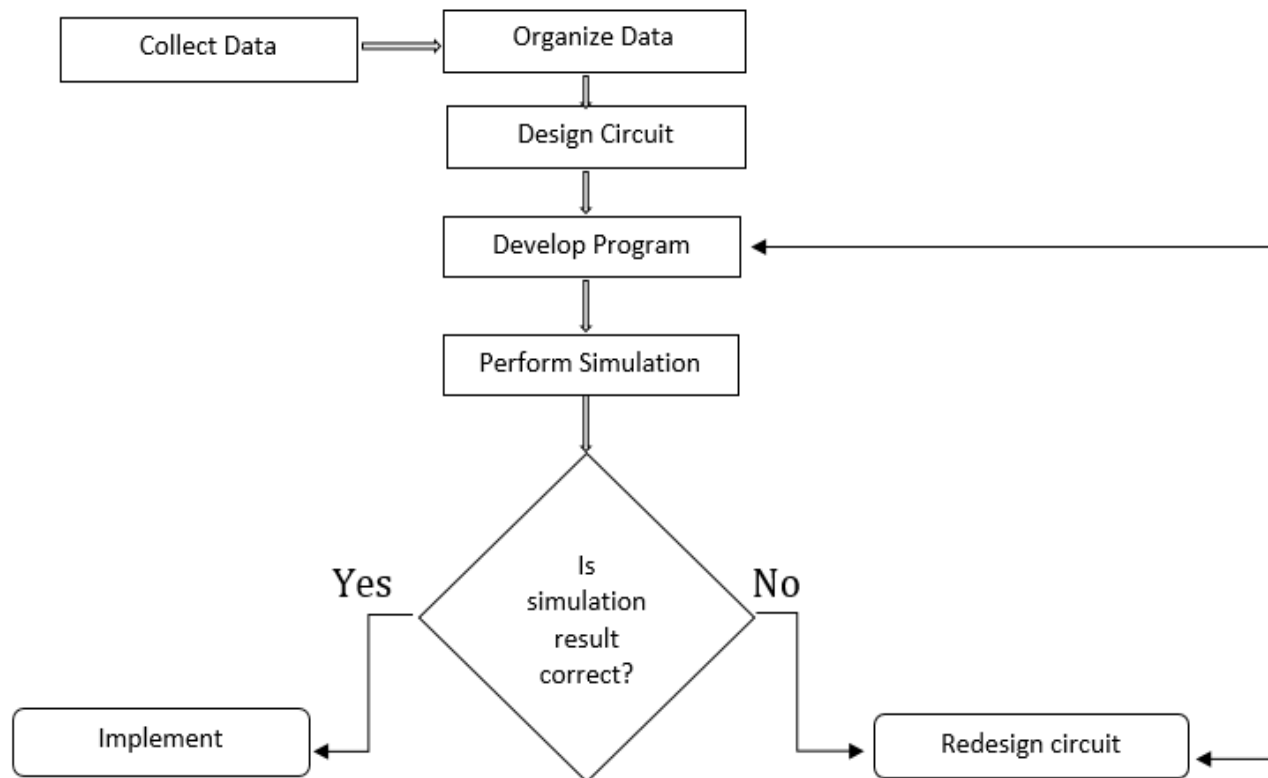


Figure 1.1: systematic diagram for methodology

1.8 Organization of the Project

This final thesis is composed of five chapters covering introduction, literature review, methodology, analysis and result and the last chapter is a discussion and conclusion.

Chapter 1: explains introduction of the focus study of the project where the number of home security has increased significantly an overview of the project.

Chapter 2: describes the literature review that we have used in this thesis.

Chapter 3: explains the specific methodology for every project scope and objective are elaborated by presenting and emphasizing the details of methods applied. Here, the block diagrams and flowcharts related to each objective or scope are also elaborated and revealed.

Chapter 4: gives every detail of the results based on the experiments and testing implemented in Chapter 3; as analyzed.

Chapter 5: presents the overall discussion on the results obtained and comparison can be done.

Here, the overall conclusion of development of the project is also enclosed together with suggestion and recommendation for future work or enhancement

CHAPTER TWO

Literature Review

2.1 General Description for Home Security Systems

Security systems are vital for protection of information and property which it is necessary for prevention them from theft or crime. The simplicity of design makes it compatible in different scenarios such as a home or commercial security system like datacenters and banks. In this experimental work it has been programmed the microcontroller AT89C51, and connected it with other electronic components to economize a security of money cabinet. The assembly language was used to program this microcontroller to control for open and close operations of the money cabinet by using limited keys as password of known person. The alarm light emitting diode LED would light by pressing the false key then the camera would take the picture for this unknown person.

Security and safety is one of the most talked of topics in almost every facet like surveillance, industrial applications, offices, and in general, in smart environments. To secure it against theft, crime, fire, etc. a powerful security system is required not only to detect but also pre-empt hazards. Conventional security systems use cameras and process large amounts of data to extract features with high cost and hence require significant infrastructures.

The integrated circuit technology is the one of the most important fields of the electronics since it minimizes the area of the circuit, parasitic effects and the cost. Furthermore, these devices are more reliable than the circuits that are set up of the discrete components. Therefore, the interest in this subject is increasing day by day. In topic, a general-purpose 8-bit microcontroller is designed using Very Large Scale Integrated Circuit (VLSI) design methods.

Each business owner attempts to protect their assets, employees and office space. We have worked to develop and maintain our businesses. Whenever one leaves, one wishes to ensure that everything is secured from malicious attacks. There are several techniques to help improve security levels at a business firm. One of the most efficient solutions is installing a security system that is self-monitored.

Security contact is an easy and reliable form of security that could be installed swiftly. These security contacts will trigger an alarm once an intruder opens any window or door whenever the system remains activated. All those contacts get installed on each connection point at doors and windows. When they are in contact, the alarm will not go off, but it will when they unexpectedly lose contact.

The purpose of motion Sensors is to detect intruders, whether inside the premises or outside. One can connect those to either the lights that work outside the building or to one alarm so that the alarm will alert you whenever an intruder lurks into the premises. Special motion

detectors allow facilities with respect to pets. Most importantly, these motion sensors are able to be activated or deactivated as and when required– which means one can save electricity or avoid unnecessary alarm alerts by switching them off during the day time, and turn them on again during the night time when danger is most.

There are many experimental researches for the security system using microcontroller, it will be mentioned some of them as the following:

In 2009 , Attaullah Khawaja designed low cost electronic system which can control different locking mechanisms. Hardware design of an electromechanical lock is also presented. Microchip Peripheral Interface Controller (PIC) microcontroller is the brain controlling all the operations. Low operating power consumption, small circuit, non-volatile memory operation using on-chip Electrically Erasable Programmable Read Only Memory (EEPROM), flexible mechanical design and user friendly interface are the salient features of this security system.

In 2011, Zamshed Iqbal Chowdhury et.al. Developed of a low-cost security system using small (Pyro electric Infrared) PIR sensor built around a microcontroller. The low-power PIR detectors take advantage of pyro electricity to detect a human body that is a constant source of Passive Infrared (radiation in the infrared region). The system sensed the signal generated by PIR sensor detecting the presence of individuals not at thermal equilibrium with the surrounding environment. Detected the presence of any unauthorized person in any specific time interval, it triggers an alarm & sets up a call to a predefined number through a global system mobile (GSM) modem. This highly reactive approach has low computational requirement, therefore it is well-suited to surveillance, industrial applications and smart environments. Tests performed gave promising results.

In 2011, Rajesh Kannan Megalingam, et.al. Deployed the home security system at the homes where elders live. This can also be installed at all homes where there is threat from robbery. This system automatically sent a short message system (SMS) or calls the police if the doors or windows are opened by an external person. The door breakage module informs the General Packet Radio Service (GPRS) modem about the theft attempt through a wireless interface. The GPRS modem calls for necessary help spotting the location. Through Person Identification Function (PIDF) which is part of Intelligent Wireless Home Security System (IWHSS), the elders can monitor the people outside the house and also talk with them through the intercom provided. Through this cost effective setup, the life of many elders can be saved even though they are living alone and their lives will not be in anymore danger.

In 2011, Xizi Li and Changyi Jiapresented an industrial temperature measuring and monitoring system, which is based on 89C51single chip microcontroller, and details the comparison between Formula method and Look-up Table method whose new algorithm and high precision has been emphasized. The system can measure real-time industrial temperature, store historical data and display temperature in digital and curve diagram.max6675 is applied

in temperature acquisition module, which simplifies the system design and improves measuring precision.

In 2012, R. Ramaniet. Al. Designed and implemented a bank locker security system based on Radio-frequency identification (RFID) and Global System Mobile (GSM) technology which can be organized in bank, secured offices and homes. In this system only authentic person can be recovered money from bank locker. It had implemented a bank locker security system based on RFID and GSM technology containing door locking system using RFID and GSM which can activate, authenticate, and validate the user and unlock the door in real time for bank locker secure access. The main advantage of using passive RFID and GSM is more secure than other systems. This system consists of microcontroller, RFID reader, GSM modem, keyboard, and LCD, in this system The RFID reader reads the identified (id) number from passive tag and send to the microcontroller, if the Id number is valid then microcontroller send the SMS request to the authenticated person mobile number, for the original password to open the bank locker, if the person send the password to the microcontroller, which will verify the passwords entered by the key board and received from authenticated mobile phone. If these two passwords are matched the locker will be opened otherwise it will be remain in locked position, this system is more secure than other systems because two passwords required for verification. This system also creates a log containing check-in and check-out of each user along with basic information of user.

2.2 Microcontrollers

Microprocessors and microcontrollers are widely used in embedded systems products. Microcontroller is a programmable device. A microcontroller has a CPU in addition to a fixed amount of RAM, ROM, I/O ports and a timer embedded all on a single chip. The fixed amount of on-chip ROM, RAM and number of I/O ports in microcontrollers makes them ideal for many applications in which cost and space are critical.

The Intel 8051 is a Harvard architecture, single chip microcontroller (μC) which was developed by Intel in 1980 for use in embedded systems. It was popular in the 1980s and early 1990s, but today it has largely been superseded by a vast range of enhanced devices with 8051-compatible processor cores that are manufactured by more than 20 independent manufacturers including Atmel, Infineon Technologies and Maxim Integrated Products.

8051 is an 8-bit processor, meaning that the CPU can work on only 8 bits of data at a time. Data larger than 8 bits has to be broken into 8-bit pieces to be processed by the CPU. 8051 is available in different memory types such as UV-EPROM, Flash and NV-RAM. The present project is implemented on Keil U vision. In order to program the device, Preload tool has been used to burn the program onto the microcontroller.

2.3 PIR Sensor

The PIR (Passive Infra-Red) sensor is nothing but a pyro electric device that identifies motion by recording alterations in the evidenced infrared levels that are emitted from the surrounding objects. The various forms of motion can be easily detected by checking on a signal I/O pin on a high signal. PIR sensor is an electronic device which is used in some security systems to detect an infrared emitting source . All living beings whose temperature is anything above absolute zero (-273.15°C or -459.67°F), do emit infrared radiation. This form of energy is completely invisible to the human eye but it can readily be observed by electronic devices that are designed for this purpose. The term used is “passive” in such an instance since PIR does not emit any energy. It merely sits passively and accepts infrared energy waves through windows in the house or industry.

The Passive Infrared PIR Sensor modules are moderately priced and now available in the market. Upon using such pre-configured and readymade PIR sensors, any average mind will be able to construct own Motion Sensor Unit.

2.4 EEPROM

EEPROM (Electrically Erasable Programmable Read only memory) has several advantages over other memory devices, such as the fact that its method of erasure is electrical and therefore instant. In addition, in EEPROM one can select which byte to be erased, in contrast to flash, in which the entire contents of ROM are erased. The main advantage of EEPROM is that one can program and erase its contents while it is in system board. It does not require physical removal of the memory chip from its socket. In general, the cost per bit for EEPROM is much higher when compared to other devices.

CHAPTER THREE

Hardware design and Methodology

3.1 Project description

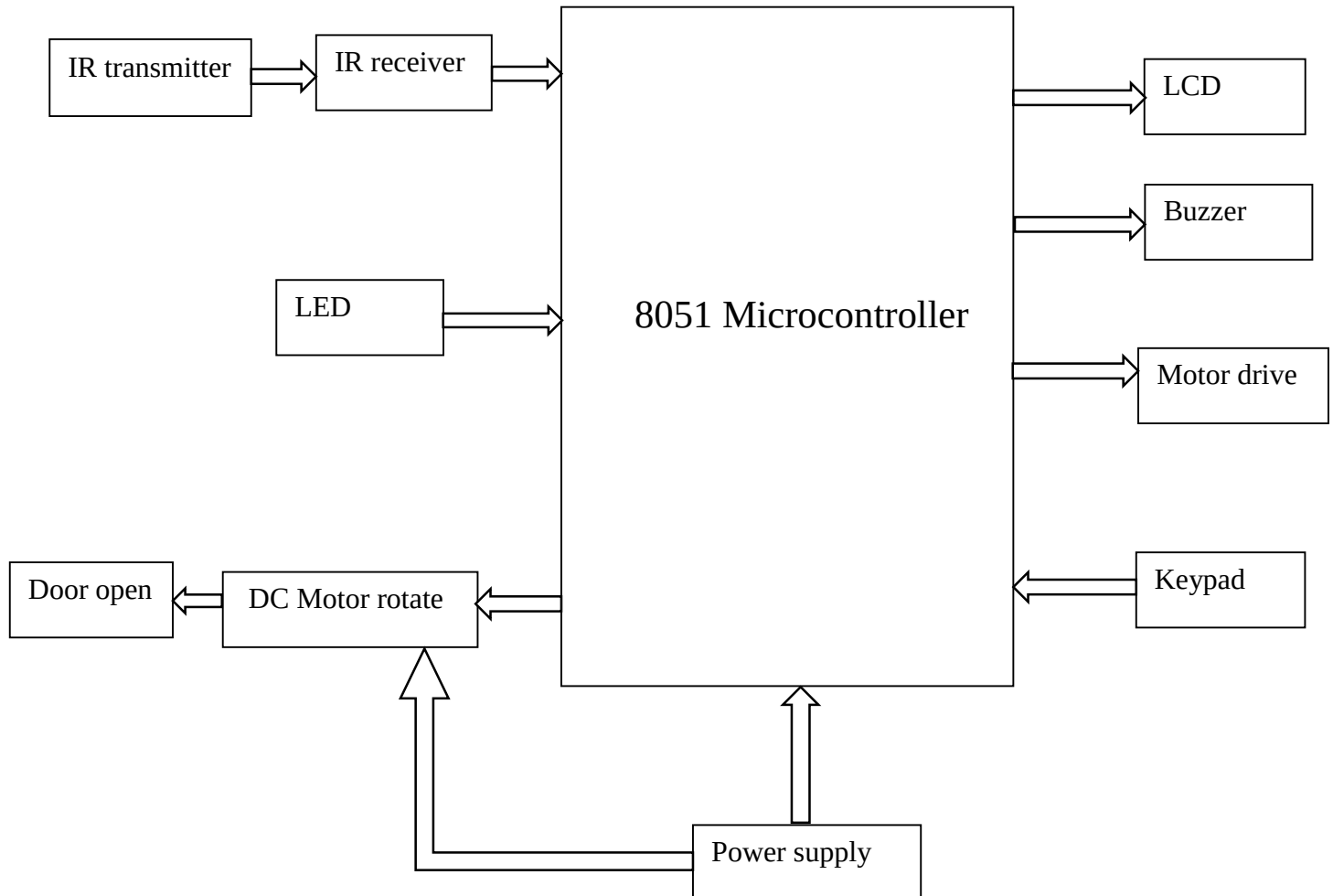


Figure 3.1: Hardware block diagram

3.2 What is the 8051 microcontroller?

The microcontroller incorporates all the features that are found in microprocessor. The microcontroller has built in ROM, RAM, Input Output ports, Serial port, timers, interrupts and

clock circuit (j.ayala). A microcontroller is an entire computer manufactured on a single chip. Microcontrollers are usually dedicated devices embedded within an application.

For example, microcontrollers are used as engine controllers in automobiles and as exposure and focus controllers in cameras.

In order to serve these applications, they have a high concentration of on-chip facilities such as serial ports, parallel input output ports, timers, counters; interrupt control, analog-to-digital converters, random access memory, read only memory, etc. The I/O, memory, and on-chip peripherals of a microcontroller are selected depending on the specifics of the target application. Since microcontrollers are powerful digital processors, the degree of control and programmability they provide significantly enhances the effectiveness of the application.

The 8051 is the first microcontroller of the MCS-51 family introduced by Intel Corporation at the end of the 1970s. The 8051 family with its many enhanced members enjoys the largest market share, estimated to be about 40%, among the various microcontrollers Architectures. The microcontroller has on chip peripheral devices. In this unit firstly we differentiate microcontroller from microprocessor then we will discuss about Hardware details of 8051 and then introduce the Assembly level language in brief.

3.2.1 Features of 8051 microcontroller

- 4K Bytes of In-System Programmable
- 4.0V to 5.5V Operating Range
- Fully Static Operation: 0 Hz to 33 MHz
- Three-level Program Memory Lock
- 128 x 8-bit Internal RAM
- 32 Programmable I/O Lines
- Two 16-bit Timer/Counters
- Six Interrupt Source

3.2.2 Description of the 8051 microcontroller

The AT89c51 is a low-power, high-performance CMOS 8-bit microcontroller with 4KBytes of in-system programmable Flash memory. The device is manufactured using Atmel's high-density nonvolatile memory technology and is compatible with the industry-standard 80C51 instruction set and pin out. The on-chip Flash allows the program memory to be reprogrammed in-system or by a conventional nonvolatile memory programmer. By combining a versatile 8-bit CPU with in system programmable Flash on a monolithic chip, the Atmel AT89c51 is a powerful microcontroller which provides a highly-flexible and cost-effective solution to many embedded control applications.

The AT89c51 provides the following standard features: 4K bytes of flash, 128 bytes of RAM, 32 I/O lines, Watchdog timer, two data pointers, two 16-bit timer/counters, and a five vector Two

level interrupt architecture, a full duplex serial port, on-chip oscillator, and clock circuitry. In addition, the AT89C51 is designed with static logic for operation down to zero frequency and supports two software selectable power saving modes.

The Idle Mode stops the CPU while allowing the RAM, timer/counters, serial port, and interrupt system to continue functioning. The Power-down mode saves the RAM contents but freezes the oscillator, disabling all other chip functions until the next external interrupt or hardware reset.

3.3 Pin configuration

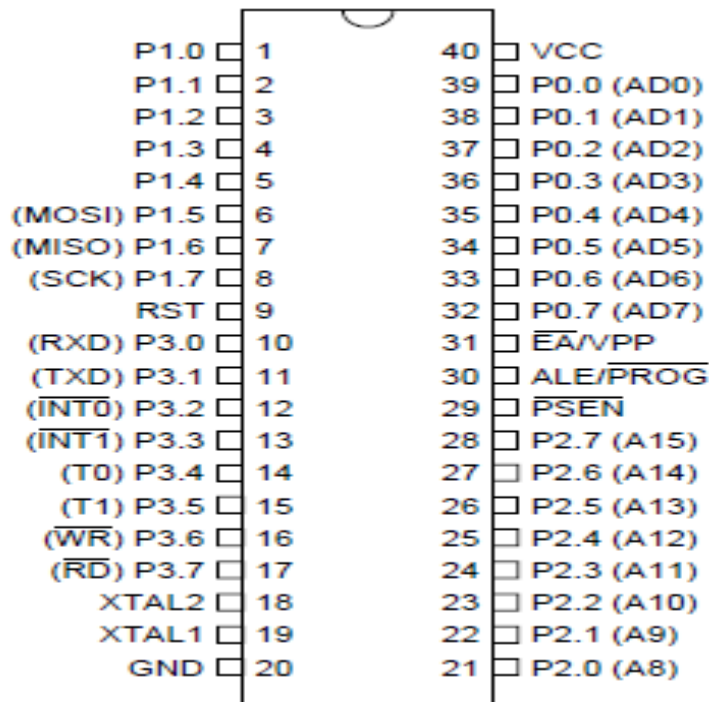


Figure 3.2: Pin description of 8051 microcontroller

VCC is the Supply voltage.

GND is Ground.

Port 0: Port 0 is an 8-bit open drain bidirectional I/O port. As an output port, each pin can sink eight TTL inputs. When 1s are written to port 0 pins, the pins can be used as high-impedance inputs. Port 0 can also be configured to be the multiplexed low-order address/data bus during accesses to external program and data memory. In this mode, P0 has internal pull-ups. Port 0 also receives the code bytes during Flash programming and outputs the code bytes during program verification. External pull-ups are required during program verification.

Port 1: Port 1 is an 8-bit bidirectional I/O port with internal pull-ups. The Port 1 output buffers can Sink/source inputs. When 1s are written to Port 1 pins, they are pulled high by the internal pull-ups and can be used as inputs. As inputs, Port 1 pins that are externally being Pulled low will

source current because of the internal pull-ups. Port 1 also receives the low-order address bytes during Flash programming and verification.

Port 2: Port 2 is an 8-bit bidirectional I/O port with internal pull-ups. The Port 2 output buffers can sink/source four TTL inputs. When 1s are written to Port 2 pins, they are pulled high by the internal pull-ups and can be used as inputs. As inputs, Port 2 pins that are externally being Pulled low will source current (IIL) because of the internal pull-ups. Port 2 emits the high-order address byte during fetches from external program memory and during accesses to external data memory that uses 16-bit addresses (MOVX @ DPTR). In this application, Port 2 uses strong internal pull-ups when emitting 1s. During accesses to external data memory that uses 8-bit addresses (MOVX @ RI) Port 2 emits the contents of the P2 Special Function Register. Port 2 also receives the high-order address bits and some control signals during Flash programming and verification.

Port 3: Port 3 is an 8-bit bidirectional I/O port with internal pull-ups. The Port 3 output buffers can sink/source four TTL inputs. When 1s are written to Port 3 pins, they are pulled high by the internal pull-ups and can be used as inputs. As inputs, Port 3 pins that are externally being Pulled low will source current (IIL) because of the pull-ups. Port 3 receives some control signals for Flash programming and verification. Port 3 also serves the functions of various special features of the AT89C51.

RST: Reset input. A high on this pin for two machine cycles while the oscillator is running resets the device. This pin drives High for 98 oscillator periods after the Watchdog times out. The DISRTO bit in SFR AUXR (address 8EH) can be used to disable this feature. In the default state of bit DISRTO, the RESET HIGH out feature is enabled.

ALE/PROG: Address Latch Enable (ALE) is an output pulse for latching the low byte of the address during accesses to external memory. This pin is also the program pulse input (PROG) during Flash programming. In normal operation, ALE is emitted at a constant rate of 1/6 the oscillator frequency and may be used for external timing or clocking purposes. Note, however, that one ALE pulse is skipped during each access to external data memory.

If desired, ALE operation can be disabled by setting bit 0 of SFR location 8EH. With the bit set, ALE is active only during a MOVX or MOVC instruction. Otherwise, the pin is weakly pulled high. Setting the ALE-disable bit has no effect if the microcontroller is in external execution mode.

PSEN: Program Store Enable (PSEN) is the read strobe to external program memory. When the AT89S51 is executing code from external program memory, PSEN is activated twice each machine cycle, except that two PSEN activations are skipped during each access to external data memory.

EA/VPP: External access enables. EA must be strapped to GND in order to enable the device to fetch code from external program memory locations starting at 0000H up to FFFFH. Note, however, that if lock bit 1 is programmed, EA will be internally latched on reset.

EA should be strapped to VCC for internal program executions. This pin also receives the 12- volt programming enable voltage (VPP) during Flash programming.

XTAL1 Input to the inverting oscillator amplifier and input to the internal clock operating circuit.

XTAL2 Output from the inverting oscillator amplifier.

3.4 LCD (liquid crystal display)

3.4.1 Description of LCD

LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications. In this project we use the 16x2 LCD display; it's a very basic module and is very commonly used in various devices and circuits. These modules are preferred over seven segments and other multi segment LEDs. LCDs are economical; easily programmable; have no limitation of displaying special & even custom characters (unlike in seven segments), animations and so on.

A 16x2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data. The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc.

3.4.2 LCD Commands and Instruction set

Only the instruction register (IR) and the data register (DR) of the LCD can be controlled by the MCU. Before starting the internal operation of the LCD, control information is temporarily stored into these registers to allow interfacing with various MCUs, which operate at different speeds, or various peripheral control devices. The internal operation of the LCD is determined by signals sent from the MCU.

These signals, which include register selection signal (RS), read/write signal (R/W), and the data bus (DB0 to DB7), make up the LCD instructions.

There are four categories of instructions that:

- Designate LCD functions, such as display format, data length, etc.
- Set internal RAM addresses
- Perform data transfer with internal RAM
- Perform miscellaneous functions

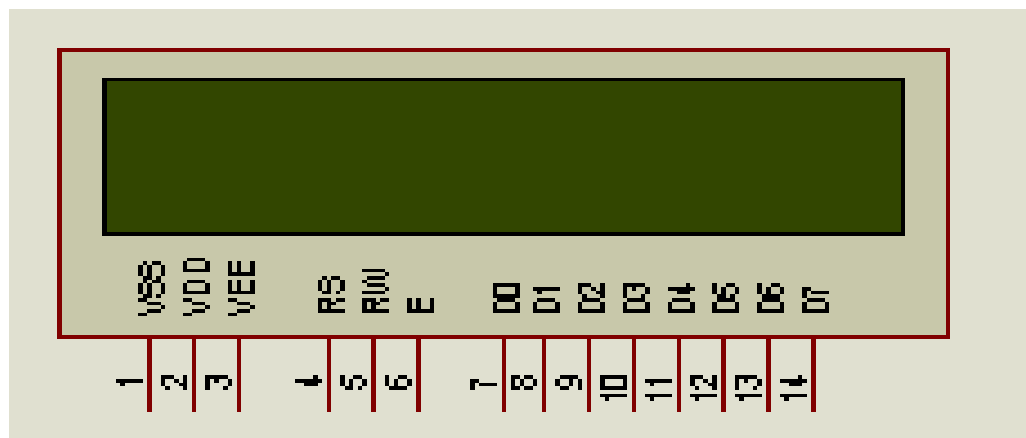


Figure 3.3: LCD pin diagram

LCD PIN description: The LCD Pin description is illustrates in the table below

Pin NO.	Name	Function
1	GROUND	Ground (0V)
2	VCC	Supply Voltage(5V)
3	VEE	Contrast adjustment through variable register
4	Register Select(RS)	(Register Select) Used to tell LCD controller weather data on Data Bus Is a Command or Text Data. High Signal for Data and Low signal for command.
5	Read/Write (R/W)	(Read Write) Used to tell LCD controller, if we wish to Read or Write To the LCD. High for Read, Low for Write.
6	Enable(EN)	Send data to data pins when high to low pulse is given
7	DB0	8-bit data pins
8	DB1	8-bit data pins
9	DB2	8-bit data pins
10	DB3	8-bit data pins
11	DB4	8-bit data pins
12	DB5	8-bit data pins
13	DB6	8-bit data pins
14	DB7	8-bit data pins

Table 3.1: LCD PIN description

3.4.3 How 16*2 modules LCD is interfaced to 89c51 microcontroller

LCD: 16×2 Liquid Crystal Display which will display the 32 characters at a time in two rows (16 characters in one row). Each character in the display of size 5 × 7 pixel matrixes, although this matrix differs for different 16×2 LCD modules if you take JHD162A this matrix goes to 5×8. This matrix will not be same for all the 16×2 LCD modules. There are 16 pins in the LCD module, the pin configuration are given in the above.

Follow these simple steps for displaying a character or data

- E=1; enable pin should be high
- RS=1; Register select should be high
- R/W=0; Read/Write pin should be low.

To send a command to the LCD just follows these steps:

- E=1; enable pin should be high
- RS=0; Register select should be low
- R/W=1; Read/Write pin should be high.

Commands: There are some preset commands which will do a specific task in the LCD. These commands are very important for displaying data in LCD. The list of commands given below:

Command	Function
0F	For switching on LCD, blinking the cursor.
1	Clearing the screen
2	Return home
4	Decrement cursor
6	Increment cursor
E	Display on and also cursor on
80	Force cursor to beginning of the first line
C0	Force cursor to beginning of second line
38	Use two lines and 5x7 matrix
83	Cursor line 1 position 3
3C	Activate second line
0C3	Jump to second line position 3
0C1	Jump to second line position1

Table 3.2: Command to the LCD

Coming to the programming you should follow these steps:

STEP1: Initialization of LCD.

STEP2: Sending command to LCD.

STEP3: Writing the data to LCD.

Initializing LCD: To initialize LCD to the 8051 the following instruction and commands are to be embed in to the functions

- 0x38 is used for 8-bit data initialization.
- 0xFH for making LCD on and initializing the cursor.
- 0X6H for incrementing the cursor which that display another character in the LCD
- 0x1H for clearing the LCD. Command

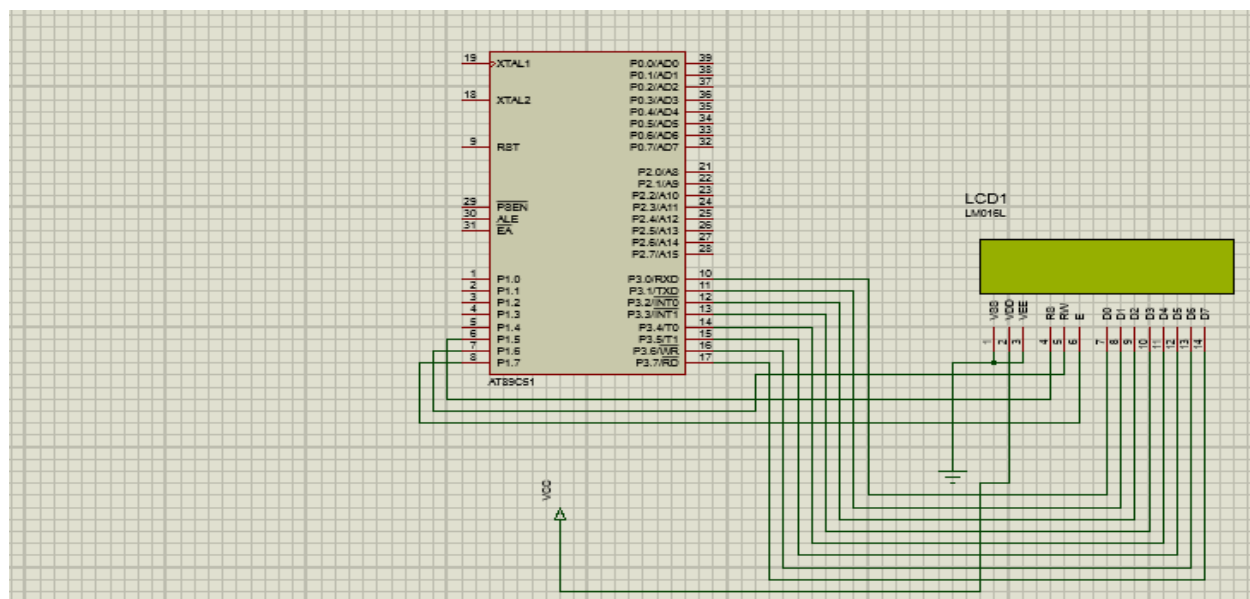


Figure 3.4: Interfacing of LCD with AT89C51

3.4.4 Sending data to the LCD

- E=1; enable pin should be high
- RS=1; Register select should be high for writing the data
- Placing the data on the data registers
- R/W=0; Read/Write pin should be low for writing the data.

3.5 Keypad

Keypad is most widely used input device to provide input from the outside world to the microcontroller. The keypad makes an application more users interactive. The concept of interfacing a keypad with the AT89c51 is similar to interfacing it with any other microcontroller. The article of interfacing keypad with 8051 can be referred for detailed description of the methodology used here. This article explains the interfacing of a 4x3 keypad with AT89c51 and

displaying the output on a LCD. The algorithm and detailed explanation for keypad interfacing is given in above mentioned article.

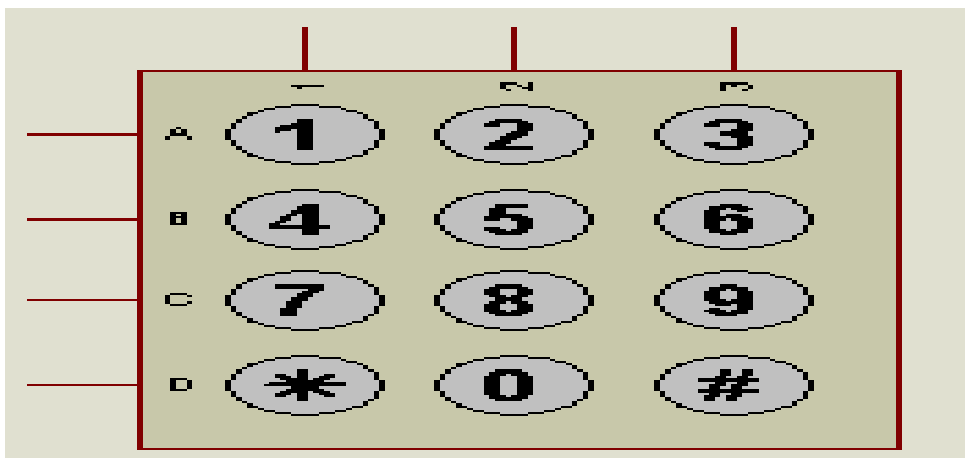


Figure 3.5: keypad

Interfacing of keypad with PIC16F887 microcontroller

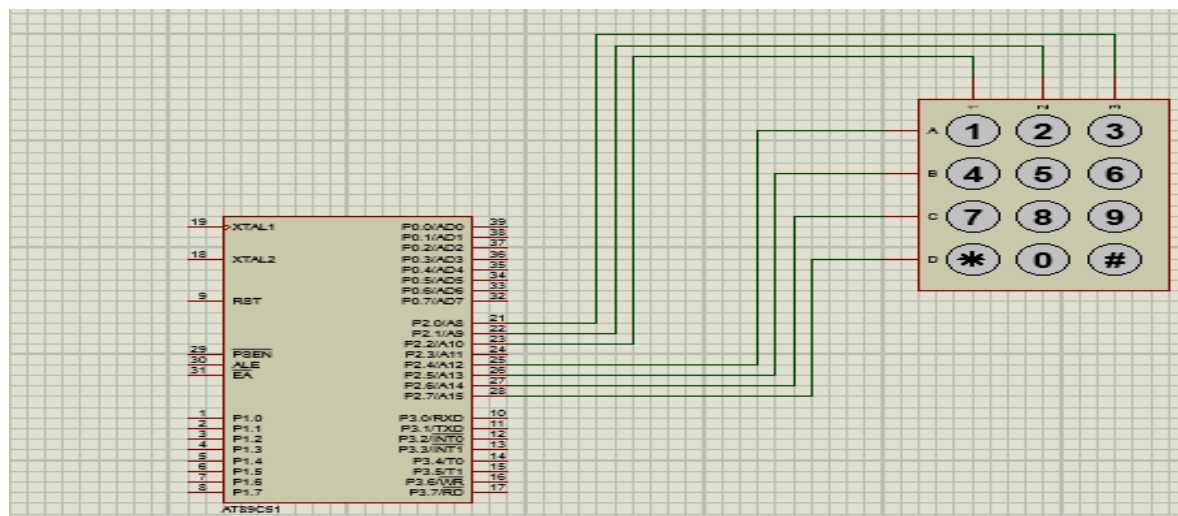


Figure 3.6: interfacing of keypad with AT89C51 microcontroller

3.6 LED (light emitting diode)

Short for Light-Emitting Diode, LED is a special semiconductor that illuminates when an electrical charge passes through it. LEDs are commonly green or red; however can be an assortment of other colors. Below are just a few examples of how an LED could be used with a computer. LED falls within the family of P-N junction devices. The light emitting diode (LED) is a diode that will give off visible light when it is energized. In any forward biased P-N junction there is, with in the structure and primarily close to the junction, a recombination of hole and electrons. This

recombination requires that the energy possessed by the unbound free electron be transferred to another state. The process of giving off light by applying an electrical source is Called electroluminescence.



Figure 3.7: LED

LED is a component used for indication. All the functions being carried out are displayed by led. The LED is diode which glows when the current is being flown through it in forward bias condition. The LEDs are available in the round shell and also in the flat shells. The positive leg is longer than negative leg.

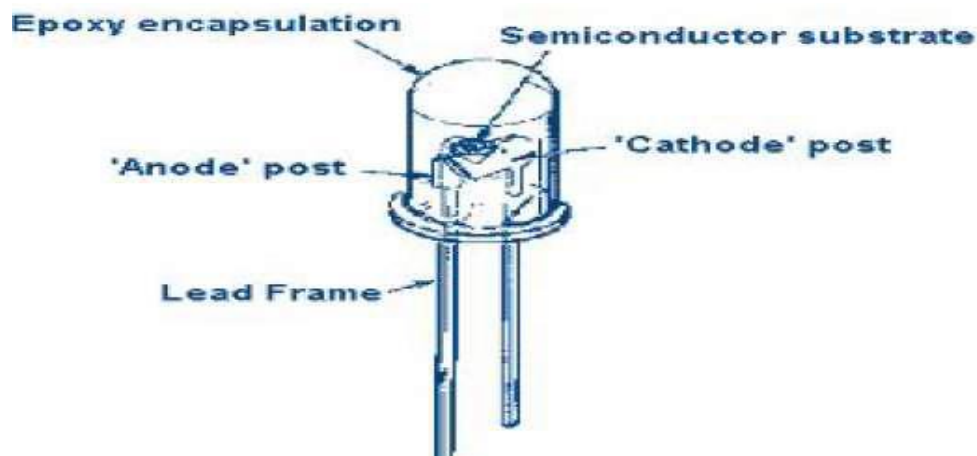


Figure 3.8: Detailed diagram of LED

3.7 Power supply

Power supply is a reference to a source of electrical power. A device or system that supplies electrical or other types of energy to an output load or group of loads is called a power supply Unit or PSU. The term is most commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others. Here in our application we need a 5v DC power supply.

For all electronics involved in the project. This requires step down transformer, rectifier, voltage regulator, and filter circuit for generation of 5v DC power.

3.8 Infrared sensor

The IR Sensor consists of an IR transmitter and an IR receiver mounted side by side on a tiny PCB. With minimum interface and 5VDC power, it can be used as a reflective type IR sensor for mobile robot or low cost object detection sensor. Now, let's take a look on the working concept of this pair of sensor Bear in mind, it is yet a complete sensor. Just we highlighted, it is just a combination of Infrared transmitter and Infrared receiver, and we will need to some interface and design to enable it become a sensor.

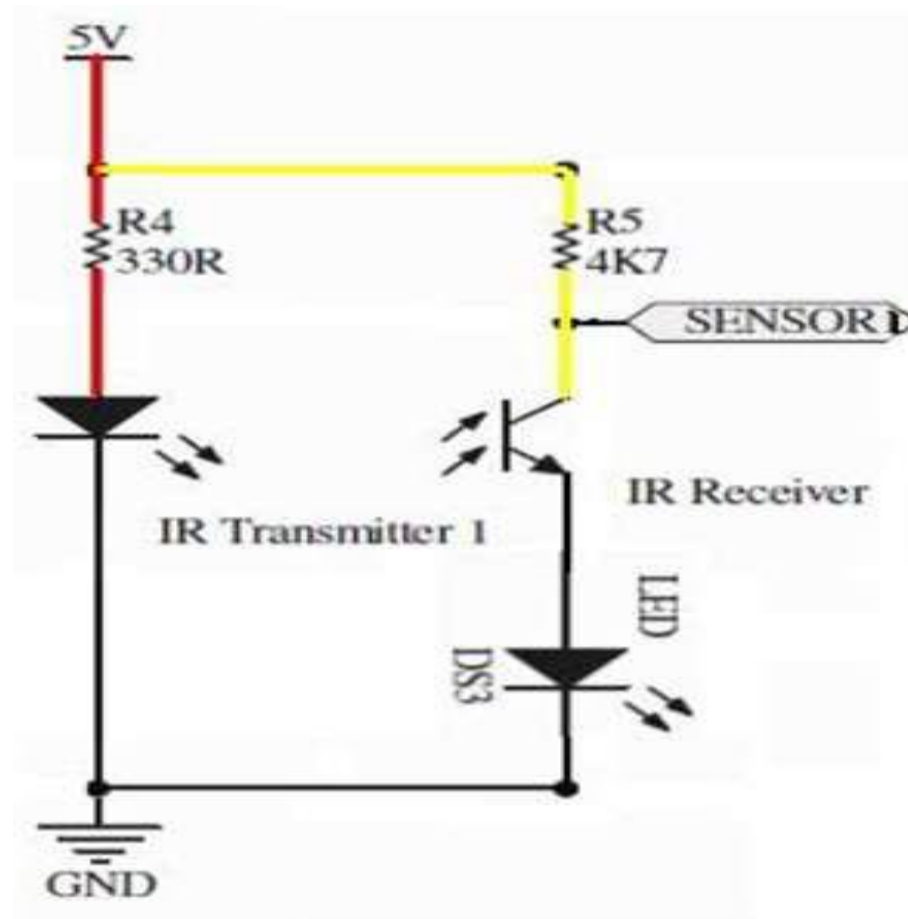


Figure 3.9: IR receiver and transmitter

SENSOR 1 located above is the part where voltage changes according to the IR received by IR Receiver. We can utilize this voltage changes to detect whether there is obstacle.

The Working Concept

IR Transmitter will always transmit IR light (Infrared), it is not visible to human eyes. Since the transmitter and receiver is being arranged side by side, theoretically, the receiver should not receive any or in most cases, it will receive small amount of infrared emitted by the IR transmitter. The working concept of IR receiver is similar to transistor or LDR (Light Dependent Resistor). Just imagine this. Referring to above diagram, the IR Receiver is like a transistor with the „base“ controlled by the IR light received. When there is no IR light receives, the „collector“ of transistor does not allow current to sink to „emitter“ further to ground of circuit. It is like very high resistance from „collector“ to „emitter“, blocking current going to ground. In this case, the voltage at sensor1 node will be high, near to 5V.

When the IR receiver receives more IR light, it changes the resistance at collector and allows more current to sink to ground, and this is similar to low resistance at the lower part of the circuit. So if you know the voltage divider formula.

3.9 AT89C51

The AT89C51 is a low-power, high-performance CMOS 8-bit microcomputer with 4Kbytes of Flash programmable and erasable read only memory (PEROM). The device is manufactured using Atmel's high-density non-volatile memory technology and is compatible with the industry standard. MCS-51 instruction set and pin out. The on- chip flash allows the program memory to be reprogrammed in-system or by a conventional non-volatile memory programmer. By combining a versatile 8-bit CPU with Flash on a monolithic chip, the Atmel AT89C51 is a powerful microcomputer which provides a highly-flexible and cost-effective solution to many embedded control applications.

3.9.1 Features of AT89C51

- Compatible with MCS-51 Products
- 4K Bytes of In-System Reprogrammable Flash Memory
- Endurance: 1,000 Write/Erase Cycles
- Fully Static Operation: 0 Hz to 24 MHz
- Three-level Program Memory Lock
- 128 x 8-bit Internal RAM
- 32 Programmable I/O Lines
- Two 16-bit Timer/Counters
- Six Interrupt Sources

- Programmable Serial Channel
- Low-power Idle and Power-down Modes

Applications

- Indoor applications (no imp ambient light is present)
- To measure speed of object (moving at a very high speed)
- Industry
- Tachometers

3.10 DC motor

First of all, a DC motor is a motor that converts electrical power into mechanical power. The main difference between them and all the other motors is the way they revolve. Unlike other motors, dc motors does continuously rotate. A dc motor is a widely used device that translates electrical pulses into mechanical movement. This DC motor is used for position control in applications such as disk drives, dot matrix printers and robotics.

We can't drive a DC Motor (depends) directly with a Microcontroller, as DC Motors requires high current and high voltage than a Microcontroller can handle. Microcontrollers usually operates at +5 or +3.3V supply and it I/O pin can provide only up to 25mA current. Commonly used DC Motors requires 12V supply and 300mA current, moreover interfacing DC Motors directly with Microcontrollers may affect the working of Microcontroller due to the Back EMF of the DC Motor. Thus it is clear that, it not a good idea to interface DC Motor directly with Microcontrollers. Each step is a fraction of a full circle. This fraction depends mostly from the mechanical parts of the motor, and from the driving method. The stepper motors also differs in the way they are powered. Instead of an AC or a DC voltage, they are driven (usually) with pulses. Each pulse is translated into a degree of rotation. Often, due to this characteristic, stepper motors are called also digital motors. As all motors, the dc motors consists of a stator and a rotor. The rotor carries a set of permanent magnets, and the stator has the coils.



Figure 3.10: DC motor

There are coils with 90 degree angle between each other fixed on the stator. The way that the coils are interconnected, will finally characterize the type of stepper motor connection. In the above drawing, the coils are not connected together. The above motor has 90 degree rotation step. The coils are activated in a cyclic order, one by one. The rotation direction of the shaft is determined by the order that the coils are activated

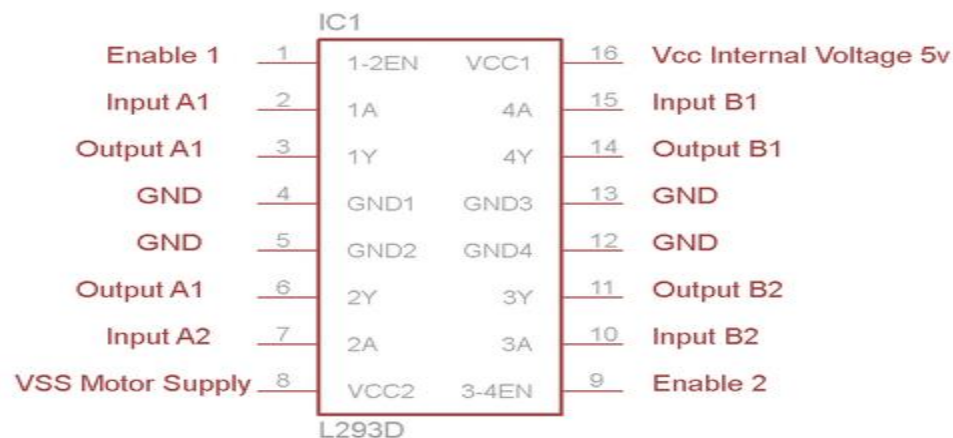
3.11 L293D Motor Driver IC

L293D is a typical Motor driver or Motor Driver IC which allows DC motor to drive on either direction. L293D is a 16-pin IC which can control a set of two DC motors simultaneously in any direction. It means that you can control two DC motor with a single L293D IC. Dual H-bridge Motor Driver integrated circuit (IC).

It works on the concept of H-bridge. H-bridge is a circuit which allows the voltage to be flown in either direction. As you know voltage need to change its direction for being able to rotate the motor in clockwise or anticlockwise direction, hence H-bridge IC are ideal for driving a DC motor.

In a single l293d chip there two h-Bridge circuit inside the IC which can rotate two dc motor independently. Due its size it is very much used in robotic application for controlling DC motors. Given below is the pin diagram of a L293D motor controller.

There are two Enable pins on l293d. Pin 1 and pin 9, for being able to drive the motor, the pin 1 and 9 need to be high. For driving the motor with left H-bridge you need to enable pin 1 to high. And for right H-Bridge you need to make the pin 9 to high. If anyone of the either pin1 or pin9 goes low then the motor in the corresponding section will suspend working. It's like a switch.



Picture 3.11: L293D Pin Diagram

Pin	description
Pin 1	Enable pin for motor 1 [Enable 1]
Pin 2	Input pin 1 for Motor 1 [Input 1]
Pin 3	Output pin 1 for Motor 1 [Output 1]
Pin 4, 5, 12, 13	Ground [GND]
Pin 6	Output Pin 2 for Motor 1 [Output 2]
Pin 7	Input pin 2 for motor 1 [Input 2]
Pin 8	Power supply for motors (9-12v) [Vcc]
Pin 9	Enable pin for motor 2 [Enable 2]
Pin 10	Input pin 1 for motor 1 [Input 3]
Pin 11	Output pin 2 for motor 1 [Output 3]
Pin 14	Output 2 for motor 1 [Output4]
Pin 15	Input 2 for motor 1 [Input 4]
Pin 16	supply voltage; 5V [Vcc1]

Table 3.3: Description of L293D

3.11.1 Working of L293D

There are 4 input pins for l293d, pin 2, 7 on the left and pin 15, 10 on the right as shown on the pin diagram. Left input pins will regulate the rotation of motor connected across left side and right input for motor on the right hand side. The motors are rotated on the basis of the inputs provided across the input pins as LOGIC 0 or LOGIC 1.

Control Signals and Motor Status

P2/IN1	P7/IN2	Motor Status
low	low	Idle [No rotation] [Hi-Impedance state]
low	high	Anticlockwise Direction
high	low	Clockwise
high	high	Idle [No rotation]

Table 3.4: motor control signal

Voltage Specification

VCC is the voltage that it needs for its own internal operation 5v; L293D will not use this voltage for driving the motor. For driving the motors it has a separate provision to provide motor supply VSS (Vsupply). L293d will use this to drive the motor. It means if you want to operate a motor at 9V then you need to provide a Supply of 9V across VSS Motor supply.

The maximum voltage for VSS motor supply is 36V. It can supply a max current of 600mA per channel. Since it can drive motors Up to 36v hence you can drive pretty big motors with this l293d. VCC pin 16 is the voltage for its own internal Operation. The maximum voltage ranges from 5v and up to 36v. Don't Exceed the Vmax Voltage of 36 volts or it will cause damage.

3.11.2 Interfacing DC Motor with 8051 using L293D

In some of the electronics projects you may want to control a DC Motor with 8051 microcontroller. The maximum current that can be sourced or sunk from a 8051 microcontroller is 15 mA at 5v. But a DC Motor need currents very much more than that and it need voltages 6v, 12v, 24v etc, depending upon the type of motor used. Another problem is that the back emf produced by the motor may affect the proper functioning of the microcontroller. Due to these reasons we can't connect a DC Motor directly to a microcontroller.

We cannot interface DC motor directly to the microcontroller, because:

1. Less voltage compatibility – simple DC motor requires minimum 9 V of power supply to run efficiently, but microcontroller pin have only 5 V CMOS logic which is not enough to drive DC motor.
2. Less current compatibility – simple DC motor requires greater than 300 mA current to produce enough flux or make a torque that runs motor efficiently, but microcontroller pins have only 1.6 mA current (here I am talking about only 8051) which is not enough to drive DC motor.
3. Back EMF – main and the greater difficulty in direct interfacing of motor is back current produced by induced EMF which may cause damage in microcontroller pin or whole microcontroller.

To overcome these problems you may use an H-Bridge using transistors. Freewheeling diodes or Clamp diodes should be used to avoid problems due to back emf. Thus it requires transistors, diodes and resistors, which may make our circuit bulky and difficult to assembly.

To overcome this problem the L293D driver IC is used. It is a Quadruple Half H-Bridge driver and it solves the problem completely. You needn't connect any transistors, resistors or diodes. We can easily control the switching of L293D using a microcontroller. There are two IC's in this category L293D and L293. L293D can provide a maximum current of 600mA from 4.5V

to 36V while L293 can provide up to 1A under the same input conditions. All inputs of these ICs are TTL compatible and clamp diodes is provided with all outputs. They are used with inductive loads such as relays solenoids, motors etc.

L293D contains four Half H Bridge drivers and are enabled in pairs. EN1 is used to enable pair 1 (IN1-OUT1, IN2-OUT2) and EN2 is used to enable pair 2 (IN3-OUT3, IN4-OUT4). We can drive two DC Motors using one L293D, but here we are using only one. You can connect second DC Motor to driver pair 2 according to your needs. The DC Motor is connected to the first pair of drivers and it is enabled by connecting EN1 to logic HIGH (5V). VSS pin is used to provide logic voltage to L293D. Here 8051 microcontroller, which works at 5v is used to control L293D, hence the logic voltage is 5. The motor supply is given to Vs pin of the L293D.

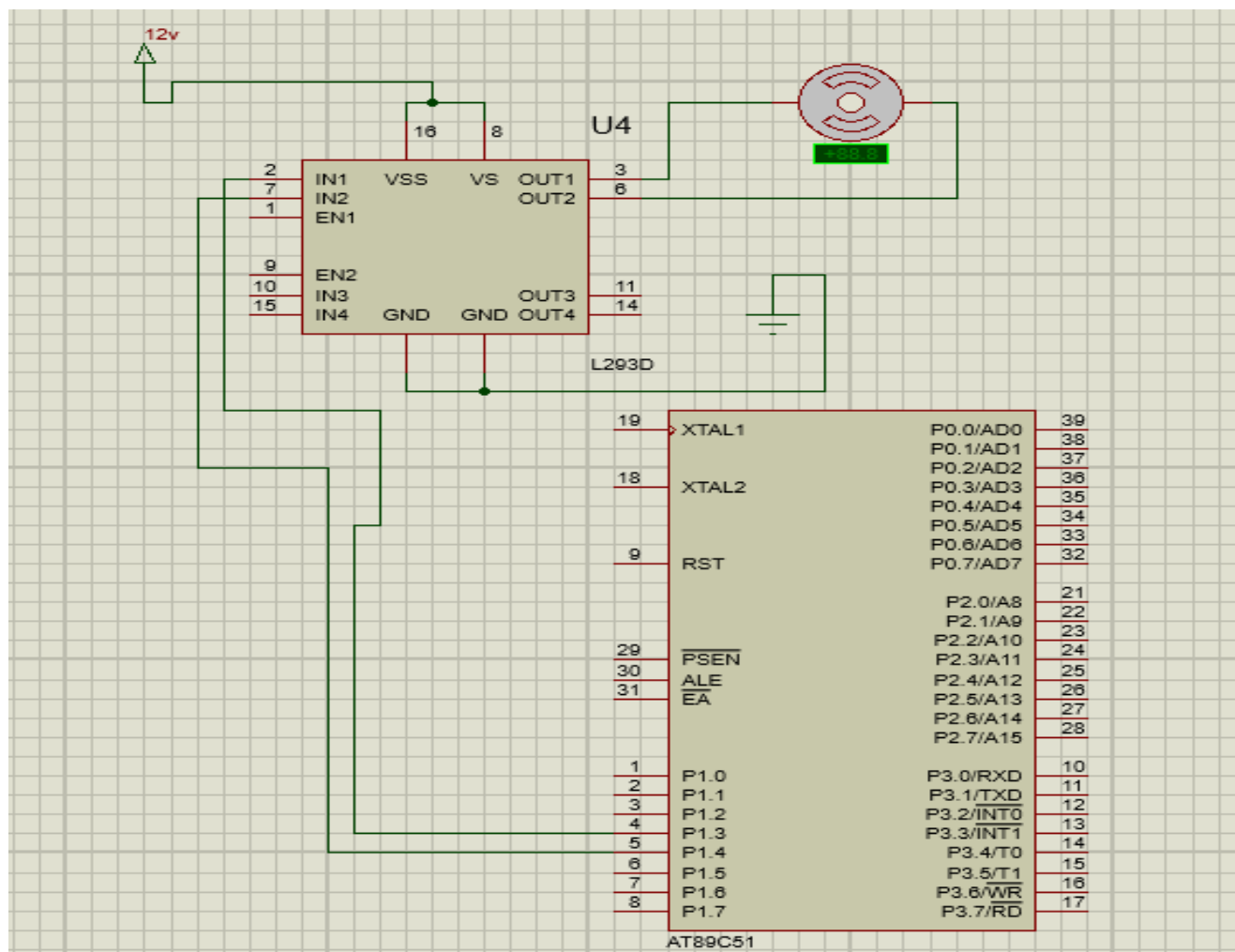


Figure 3.12: Interfacing DC Motor with 8051 using L293D

3.12 Capacitor

Capacitors are two-terminal electrical elements. Capacitors are essentially two conductors, usually conduction plates - but any two conductors - separated by an insulator - a dielectric with connection wires connected to the two conducting plates. Capacitors occur naturally. On printed circuit boards two wires running parallel to each other on opposite sides of the board form a capacitor.



Figure 3.13: capacitor

Capacitors are components that are used to store an electrical charge and are used in timer Circuits. A capacitor may be used with a resistor to produce a timer. Sometimes capacitors are used to smooth a current in a circuit as they can prevent false triggering of other components such as relays. When power is supplied to a circuit that includes a capacitor, the capacitor charges up. When power is turned off the capacitor discharges its electrical charge slowly.

3.13 Resistor

The flow of charge through any material encounters an opposing force similar in many respects to mechanical friction this opposing force is called resistance of the material .in some electric circuit resistance is deliberately introduced in form of resistor. Resistor used fall in three categories, only two of which are color coded which are metal film and carbon film resistor. The third category is the wire wound type, where value is generally printed on the vitreous paint Finish of the component. Resistors are in ohms and are represented in Greek letter omega, looks as an upturned horseshoe. Most electronic circuit requires resistors to make them work properly and it is obviously important to find out something about the different types of resistors available. Resistance is measured in ohms; the symbol for ohm is an omega ohm. 1 ohm is quite small for electronics so resistances are often given in Kohm and Mohm. Resistors used in electronics can have resistances as low as 0.1 ohm or as high as 10 Mohm.



Figure 3.14: resistor

3.14 Diode

Diode is an electronic device that allows the passage of current in only one direction. The first such devices were vacuum-tube diodes, consisting of an evacuated glass or steel envelope containing two electrodes – a cathode and an anode. The diodes commonly used in electronic circuits are semiconductor diodes. There are different diodes used in electronic circuits such as Junction diode, Zener diode, Photo diodes, and tunnel diode. Junction diodes consist of junction of two different kinds of semiconductor material. The Zener diode is a special junction type diode, using silicon, in which the voltage across the junction is independent of the current through the junction.

3.15 Crystal oscillator

A crystal oscillator is an electronic oscillator circuit that uses the mechanical resonance of a vibrating crystal of piezoelectric material to create an electrical signal with a very precise frequency. This frequency is commonly used to keep track of time (as in quartz wristwatches), to provide a stable clock signal for digital integrated circuits, and to stabilize frequencies for radio transmitters and receivers.

The most common type of piezoelectric resonator used is the quartz crystal, so oscillator circuits designed around them became known as "crystal oscillators."

3.16 EEPROM (24C32A)

In our project we use The Microchip Technology Inc. 24C32A; it is a 4K x 8 (32K bit) Serial Electrically Erasable PROM. It has been developed for advanced, low power applications such as personal communications or data acquisition. The 24C32A also has a page-write capability of up to 32 bytes of data. The 24C32A is capable of both random and sequential reads up to the 32K boundary. Functional address lines allow up to eight 24C32A devices on the same bus, for up to 256K bits address space. Advanced CMOS technology and broad voltage range make this device ideal for low-power/low-voltage, non-volatile code and data applications.

The 24C32A is available in the standard 8-pin plastic DIP and both 150 mil and 200 mil SOIC packaging. EEPROM is user-modifiable read-only memory (ROM) that can be erased and reprogrammed (written to) repeatedly through the application of higher than normal electrical voltage generated externally or internally in the case of modern EEPROMs. EPROM usually

must be removed from the device for erasing and programming, whereas EEPROMs can be programmed and erased in circuit. Originally, EEPROMs were limited to single byte operations which made them slower, but modern EEPROMs allow multi-byte page operations.

It also has a limited life - that is, the number of times it could be reprogrammed was limited to tens or hundreds of thousands of times. That limitation has been extended to a million write operations in modern EEPROMs. In an EEPROM that is frequently reprogrammed while the computer is in use, the life of the EEPROM can be an important design consideration. It is for this reason that EEPROMs were used for configuration information, rather than random access memory. Here, we are using AT 24C32A EEPROM because of that design configuration.

Features

- Voltage operating range: 4.5V to 5.5V
 - Maximum write current 3mA at 5.5V
 - Standby current 1μA typical at 5.0V
- 2-wire serial interface bus compatible
- 100 kHz and 400 kHz compatibility
- Self-timed ERASE and WRITE cycles
- Power on/off data protection circuitry
- Hardware write protect
- 1,000,000 Erase/Write cycles guaranteed
- 32-byte page or byte write modes available
- Schmitt trigger filtered inputs for noise suppression

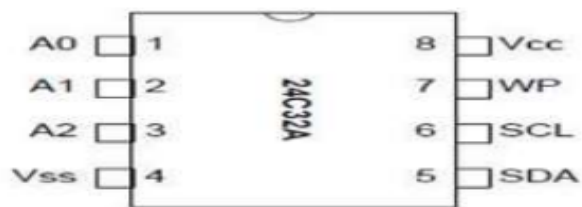


Figure 3.15: EEPROM

Pin Descriptions:

A0, A1, and A2 Chip Address Inputs: The A0, A1 and A2 inputs are used by the 24C32A for multiple device operation and conform to the 2-wire bus standard. The levels applied to these pins define the address block occupied by the device in the address map. A particular device is selected by transmitting the corresponding bits (A2, A1, and A0) in the control byte.

SDA Serial Address/Data Input/output: This is a Bi-directional pin used to transfer addresses

and data into and data out of the device. It is an open drain terminal; therefore the SDA bus requires a pull up resistor to VCC (typical 10K Ω for 100 kHz, 2 K Ω for 400 kHz) for normal data transfer SDA is allowed to change only during SCL low. Changes during SCL HIGH are reserved for indicating the START and STOP conditions. SCL Serial Clock: This input is used to synchronize the data transfer from and to the device.

WP: This pin must be connected to either VSS or VCC. If tied to VSS, normal memory operation is enabled (read/write the entire memory 000-FFF). If tied to VCC, WRITE operations are inhibited. The entire memory will be write-protected. Read operations are not affected.

Read Operation Read operations are initiated in the same way as write operations with the exception that the R/W bit of the slave address is set to one. There are three basic types of read operations: current address read, random read, and sequential read.

Current Address Read: The 24C32A contains an address counter that maintains the address of the last word accessed, internally incremented by one. Therefore, if the previous access (either a read or write operation) was to address n (n is any legal address), the next current address read operation would access data from address n + 1. Upon receipt of the slave address with R/W bit set to one, the 24C32A issues an Acknowledge and transmits the eight bit data word. The master will not acknowledge the transfer but does generate a stop condition and the 24C32A discontinues transmission.

Random Read: Random read operations allow the master to access any memory location in a random manner. To perform this type of read operation, first the word address must be set. This is done by sending the word address to the 24C32A as part of a write operation (R/W bit set to zero). After the word address is sent, the master generates a start condition following the acknowledge. This terminates the write operation, but not before the internal address pointer is set. Then the master issues the control byte again but with the R/W bit set to a one. The 24C32A will then issue an Acknowledge and transmit the 8-bit data word. The master will not acknowledge the transfer but does generate a stop condition which causes the 24C32A to discontinue transmission.

3.17 Methodology

There are several steps to be applied in designing home security system. The relevant information is gathered through literature review from previous chapter. Data on different hardware materials and security system projects has been collected where the theoretical design is studied based on microcontroller for security concept. The next is the hardware development according to the circuit designed. This process is just only being preceded if each part of the circuit being improved is valid, else, it will be repeated until it is valid as the theoretical. Once the hardware development circuits have the output as the expected, then, the comparison for both hardware and theoretical analysis will be done. Next is the step where software structure is developed for the security system to be interface with the hardware development.

In this section, we will discuss an overall overview of automatic home security system.

The goal of this project is to utilize the after-market parts and build an integrated home security system. Besides traditional magnetic switch equipped on doors and windows, we have also incorporated IR sensor to detect the blockage of the IR transmitter and IR receiver.

The control panel checks whether he/she is allowed to enter the particular door or not. If the employee is authentic, then he/she is allowed access in the particular entrance. The employees can be permitted in a given entrance as per his/her designation. The access control is employed at this point.

3.17.1 LCD Module

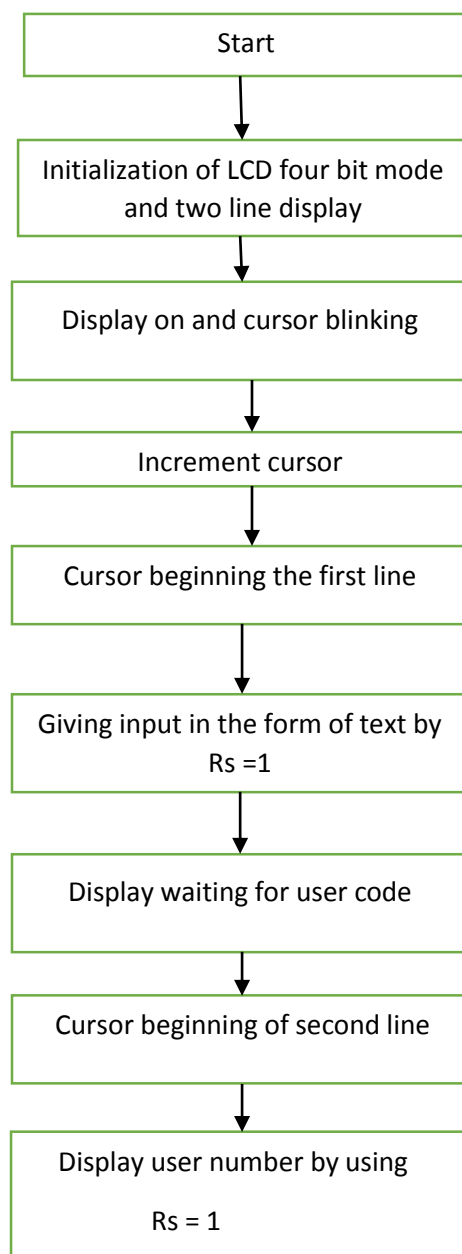


Figure 3.16: Flow chart of initialization

From the above flow chart the cursor position is settled on LCD through the following.
Means to set the cursor position on LCD, we need to send the DDRAM address

BIT	7	6	5	4	3	2	1	0
CODE	AD7	AD6	AD5	AD4	AD3	AD2	AD1	AD0

The seventh bit is always 1, and bit 0 to 7 are DDRAM address .To put the cursor on first position the address will be '0000000B' in binary and 7th bit is 1. So address will be 0x80, so for DDRAM all address starts from 0x80. For 2 lines and 16 character LCD: The address from 0x80 to 0x8F are visible on first line and 0xC0 to 0xCF are visible on second line, rest of the DDRAM area is still available but is not visible on the LCD.

Sending Data to LCD to send data simply selects the data register. Everything is same as the command routine. Following are the steps:

- ❖ Move data to LCD port
- ❖ Select data register
- ❖ Select write operation
- ❖ Send enable signal
- ❖ Wait for LCD to process the data.

3.17.2 Process flow chart

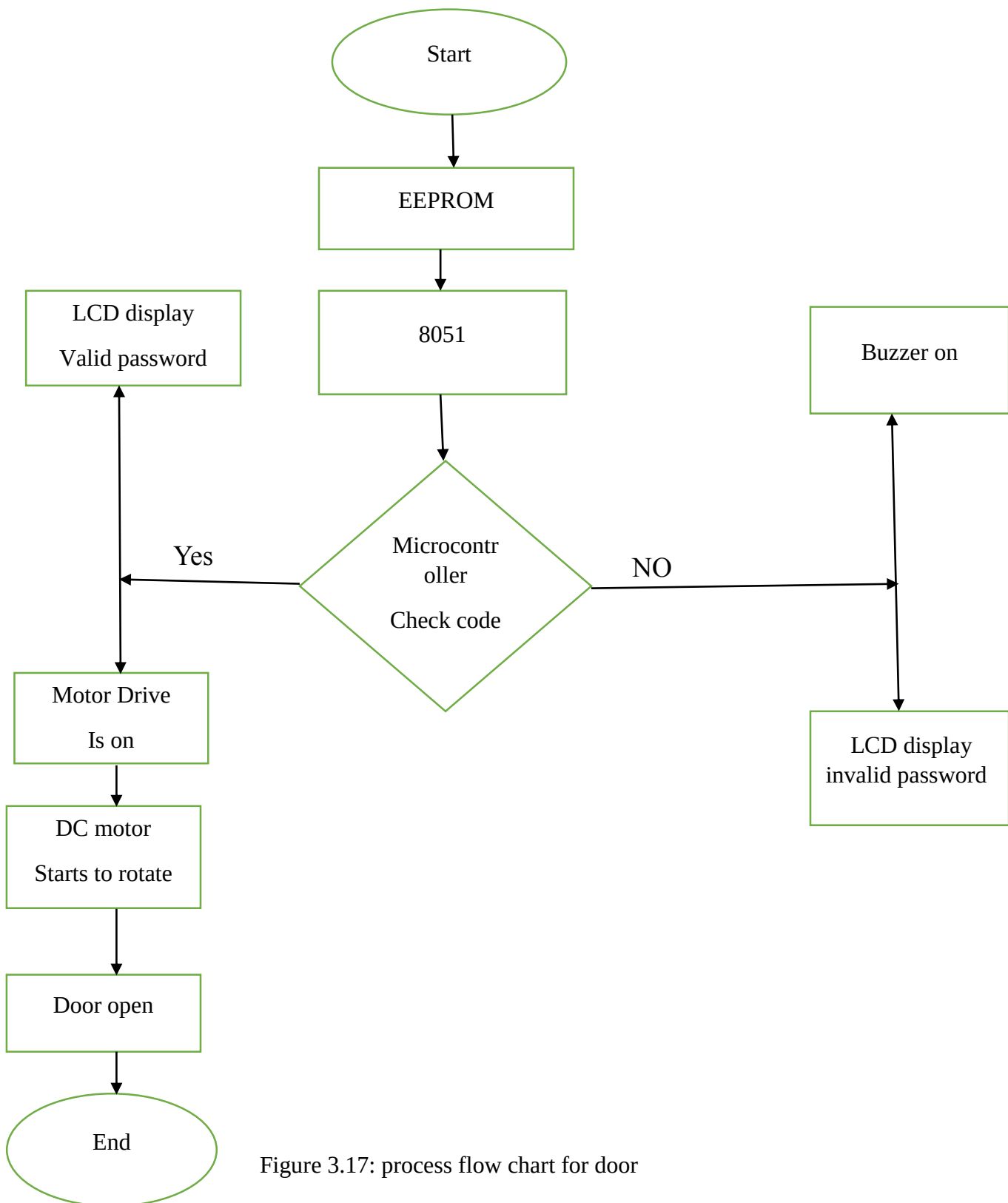


Figure 3.17: process flow chart for door

3.17.3 System design hardware interfacing for door security

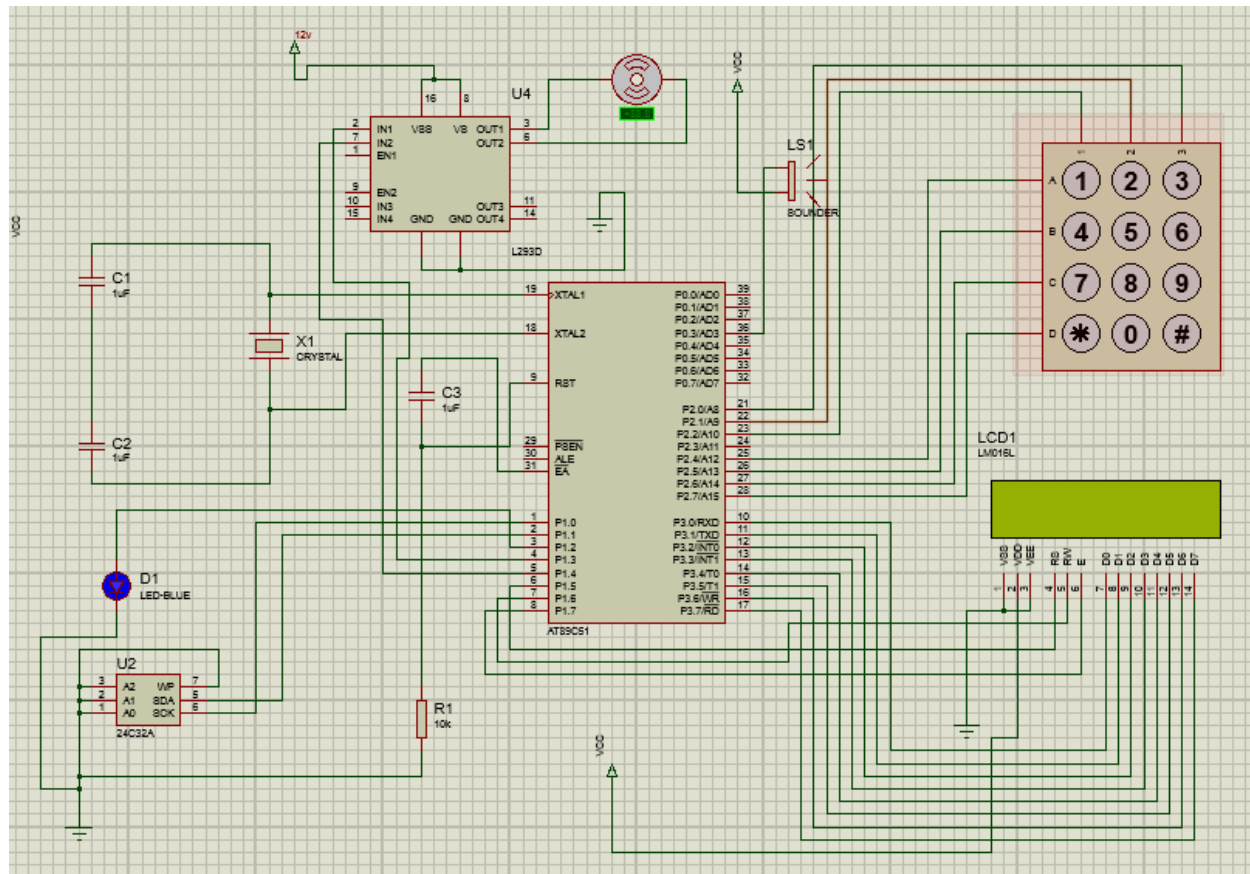


Figure 3.18: System design for door

Block diagram description for door

As we see from the above block diagram means in our design we use the port p2 as an input and the other ports p0, p1 and p3 as an output. Here both the keypad and the infrared sensor are used as an input and the buzzer and LED blue are using as an output. The keypad is connected to the port P2 of p2.0 through p2.7 except p2.3 of the Atmel 89c51 microcontroller.

The default password here is “1432”, which is stored in the EEPROM so that we can erase it at any time. As the time we entered a correct password “valid password” is displayed on LCD. Automatically the motor drive gets energized and it lets start for the stepper motor as a result the door opened. When the password entered is correct the motor begins to rotate clockwise direction. This imply that the entered password is correct and door opened or granted. After some time delay the motor begins to rotate in the reverse direction, this imply that the door is closed.

Controller receives this id then compares availability and priority of this id, if they are matched it the access to the user. If all information is matched then “valid password” message is displayed to

LCD. At the time that the entered password is not correct an “invalid password” is displayed on LCD and automatically the buzzer turned on to indicate that someone is entering wrong password and the owner may take an action.

The symbol “#” is used for two operations: it is used as ENTER to run the entered password and as they entered password has become true the symbol change its operation and used as Back to allow another password to enter.

The symbol “*” is also used for two operations; it is used to change the default password and at the same time it is used as Back at the time that the user knows that he/she entered wrong password. LCD is used in 4 bit mode, so the 8 bit data is moved in the form of nibbles to the LCD one by one. When LCD was tested by supplying it with proper location, it gave correct display according to the LCD programming with cursor on and blinking and cursor incrementing to write. LCD displays person's data base and the final result.

The project works for the door access by checking the password in EEPROM and displays the results accordingly.

3.18 Proteus

Proteus Professional design combines the ISIS schematic capture and ARES PCB layout programs to provide a powerful, integrated and easy to use tools suite for education and professional PCB design. As professional PCB design software with integrated shape based auto router, it provides features such as fully featured schematic capture, highly configurable design rules, interactive spice circuit simulator, extensive support for power planes, industry standard CAD/CAM and OD++ output, and integrated 3D viewer.

We use ISIS for simulating Intel response, it has many variety modeling libraries, and its powerful concentrates in MCUs and MPUs modeling, along with wide range of supporting chips such that AVR MCU series, 8051 MCU series, Basic stamp, HC11 MCU series, ARM CPU, Z80, Motorola 68K CPU, and most PIC's families, also it has a debugger, registers contents viewer and many other features.

All Proteus PCB design products include an integrated shape based auto router and a basic SPICE simulation capability as standard. More advanced routing modes are included in Proteus PCB Design Level 2 and higher whilst simulation capabilities can be enhanced by purchasing the Advanced Simulation option and/or micro-controller simulation capabilities.

System components:

- 1) ISIS Schematic Capture a tool for entering designs.
- 2) PROSPICE Mixed mode SPICE simulation - industry standard SPICE3F5 simulator combined with a digital simulator.

- 3) ARES PCB Layout - PCB design system with automatic component placer, rip-up and retry auto-router and interactive design rule checking.
- 4) VSM - Virtual System Modeling lets co-simulate embedded software for popular microcontrollers alongside hardware design.
- 5) System Benefits Integrated package with common user interface and fully context sensitive help.

3.19 Keil micro vision5

The Micro-Vision IDE is a Windows-based software development platform that Combines a robust editor, project manager, and makes facility. Micro-Vision integrates all Tools including the C compiler, macro assembler, linker/locator, and HEX file generator.

Keil compiler is a software used where the machine language code is written and compiled. After compilation, the machine source code is converted into hex code which is to be dumped into the microcontroller for further processing. Keil compiler also supports C language code.

It helps expedite the development process of your embedded applications by providing the following:

- ❖ Full-featured source code editor,
- ❖ Device database for configuring the development tool setting,
- ❖ Project manager for creating and maintaining your projects,
- ❖ Integrated make facility for assembling, compiling, and linking your embedded Applications,
- ❖ Dialogs for all development tool settings,
- ❖ True integrated source-level Debugger with high-speed CPU and peripheral Simulator,
- ❖ Advanced GDI interface for software debugging in the target hardware and for connection to Keil ULINK,
- ❖ Flash programming utility for downloading the application program into Flash ROM,
- ❖ Links to development tools manuals, device datasheets & user's guides.

The Micro-Vision IDE offers numerous features and advantages that help you quickly and successfully develop embedded applications. They are easy to use and are guaranteed to help you achieve your design goals.

3.20 Proload

Proload is a software which accepts only hex files. Once the machine code is converted into hex code, that hex code has to be dumped into the microcontroller and this is done by the Proload. Proload is a programmer which itself contains a microcontroller in it other than the one which is to be programmed. This microcontroller has a program in it written in such a way that it accepts the hex file from the keil compiler and dumps this hex file into the microcontroller which is to be

programmed. As the proload programmer kit requires power supply to be operated, this power supply is given from the power supply circuit. It should be noted that this programmer kit contains a power supply section in the board itself but in order to switch on that power supply, a source is required. Thus this is accomplished from the power supply board with an output of 12 volts.

Working Procedure

The Project Embedded Password based Electrical Appliances control system using 89C51 Microcontroller is an exclusive project that provides security at ATM centers, offices and homes. Security is of primary concern and in this busy, competitive world, human cannot find ways to provide security to his confidential belongings manually. Instead, he finds an alternative which can provide a full-fledged security as well as automatized. This project has been developed on this motto.

In this project every user, a part of an organization, is given a unique password. The passwords of all the users of that particular organization will be stored in EEPROM present in the system board because the data in EEPROM can be changed any number of times without the physical removal of the memory chip from its socket.

This system will be fixed at the main door of the restricted area. Whenever the user tries to enter into the organization, this system asks the user to enter his user id. This message “Enter the password” will be displayed on the LCD. Then the user has to enter his identification number. The microcontroller accepts this data and compares this user id which is already stores in the EEPROM. If the user id is not matched with any of the stored user ids, the microcontroller will not proceed for further details but displays a message “Invalid password”. If the password matches with any one of the passwords stored, the microcontroller opens the door by rotating the stepper motor through L293D motor Driver for the person to enter into the restricted area and the message “Door is opening” will be displayed on the LCD display. After some time delay the motor rotates in the reverse direction.

But if the entered password does not match with the already stored passwords, the message” invalid password” will be displayed on the LCD and the stepper motor does not rotate. At this time the buzzer will becomes activated.

CHAPTER FOUR

Result and Discussion

4.1 Result

The power supply is provided first to the hardware and through voltage regulator, the circuit components receive their proper supply voltage. After that, the LCD displays the required BCD code. The code is received by the module and checks the code availability.

When we give a power supply, the LCD will become active to accept the user password. The message “Enter the password” will be displayed on the LCD screen. At this time all components become active which is interfaced with the microcontroller.

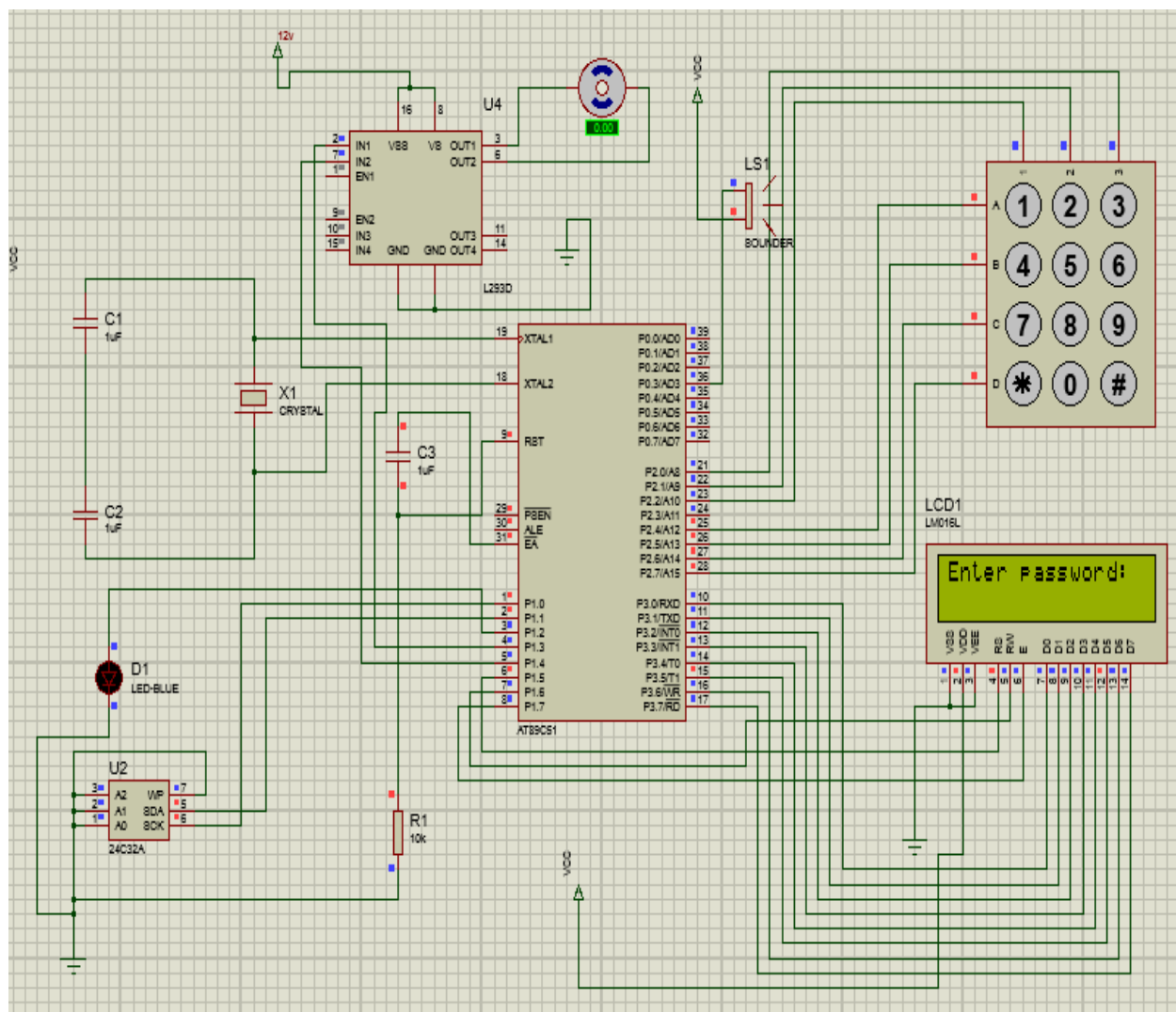


Figure 4.1 simulation result for asking to enter the user password

For correct code, LCD displays the “valid password” as follows” at same time LED turned on that we have used this LED-blue in place of IR sensor and the dc motor starts to rotate in clock wise direction to imply that door is opened. After some time delay the motor starts to rotate to the reverse direction to imply that the door is closed. When the door is closed the motor will become stopped.

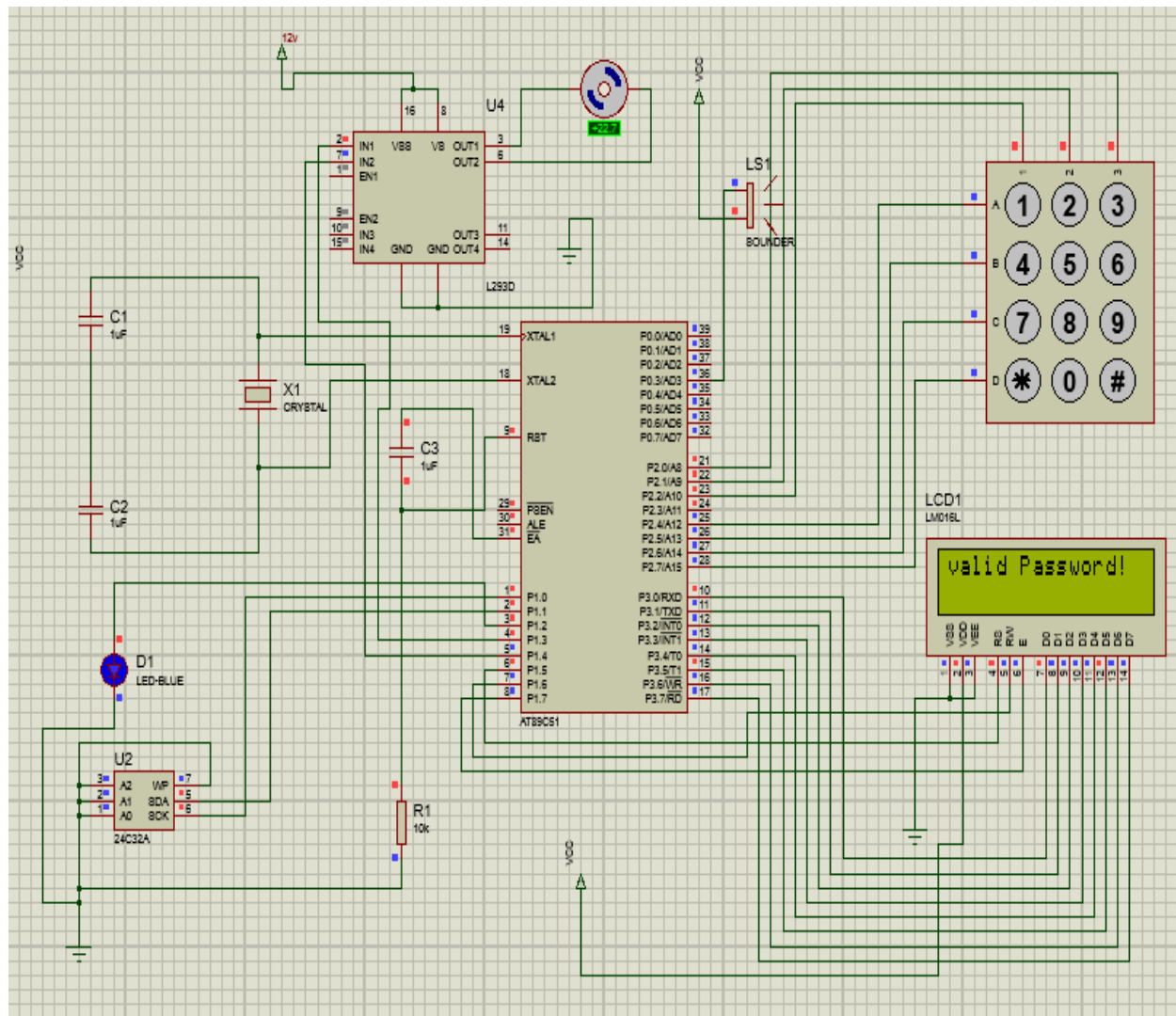


Figure 4.2: Simulation result for valid password for the door

When we enter the correct password “valid password” is displayed on LCD screen. The dc motor is rotated in clock wise direction. At this time the door must begin to open to enter the person into the house. After some time delay the message “door opening...” is displayed on the screen. This implies that the door is on the way to opened.

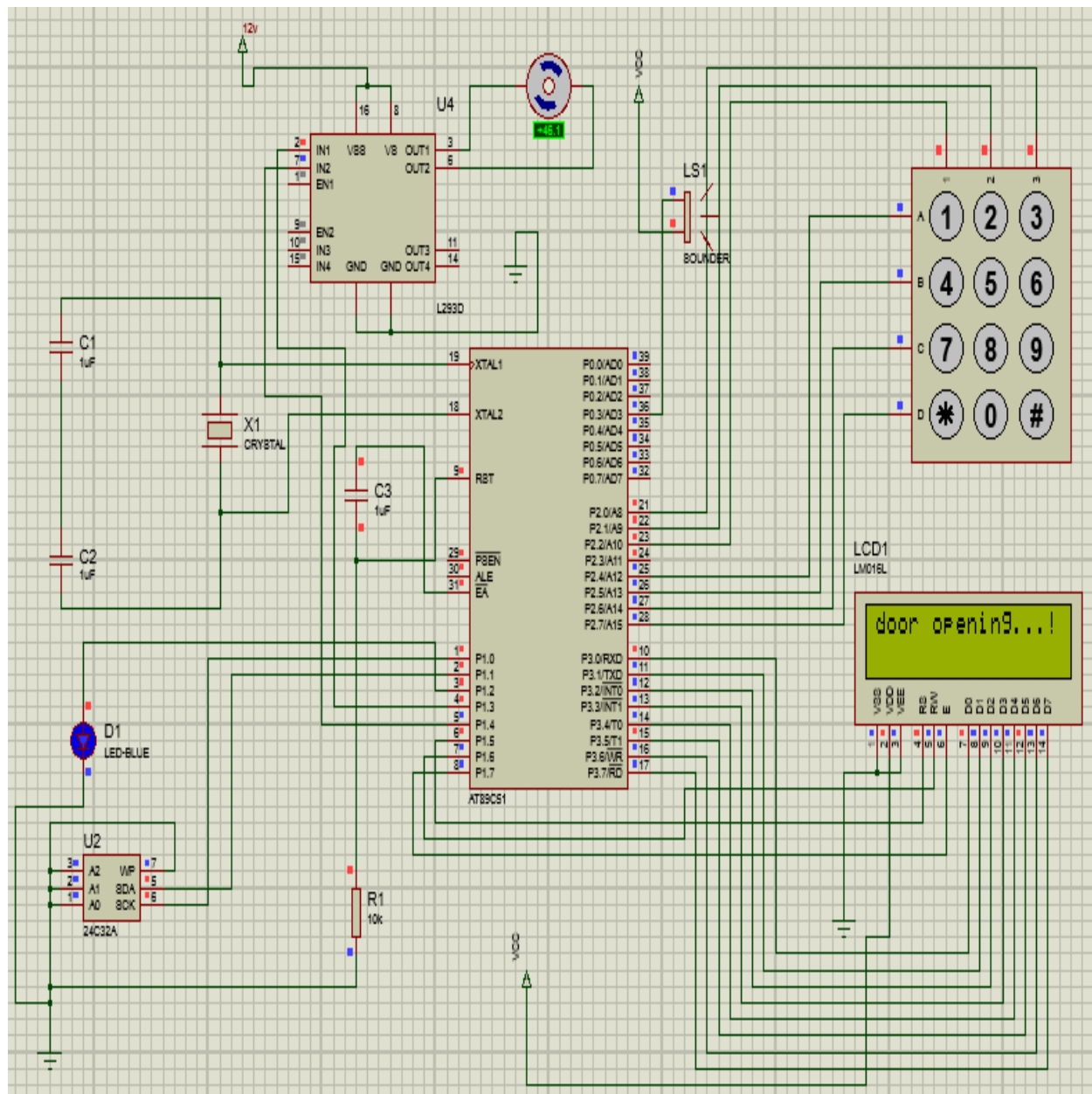


Figure 4.3 simulation result for showing that the door is opened

After valid password is entered and the door is opening, the door must be closed after all persons are entered into the home. There is a time delay for opening and closing of door. Therefore after some time delay the door will become closed. At this time the dc motor rotates in anticlockwise direction. This implies that the door is closed and a message “door closing...” is displayed on the LCD screen.

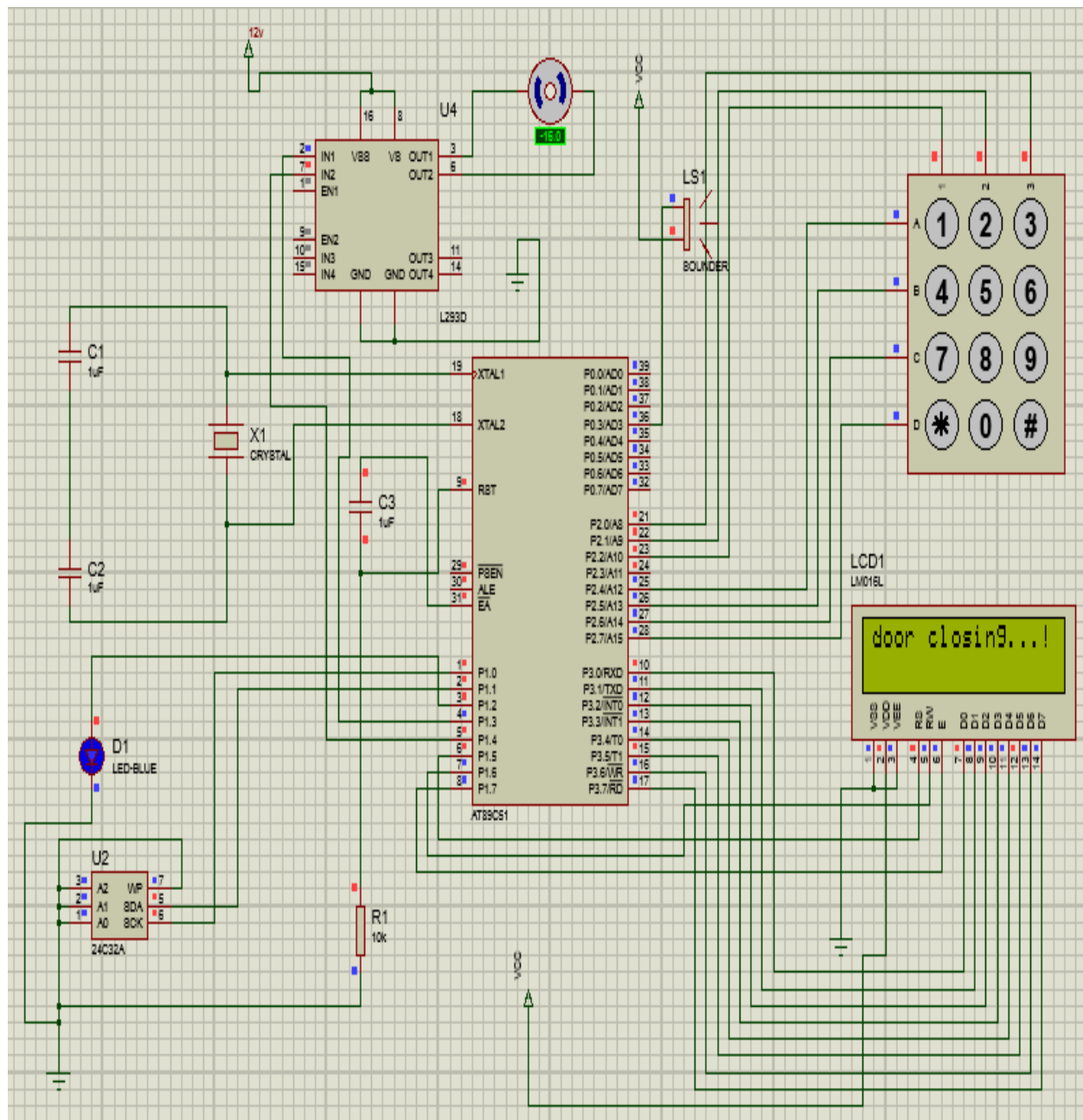


Figure 4.4 simulation result for showing the door closing

For wrong code, LCD displays “invalid password”. In this case the motor cannot be rotate. Because the motor drive L293D didn't become energized. At the same time the sounder make sound. So, the owner can take an action when unauthorized person is tried to open the door.

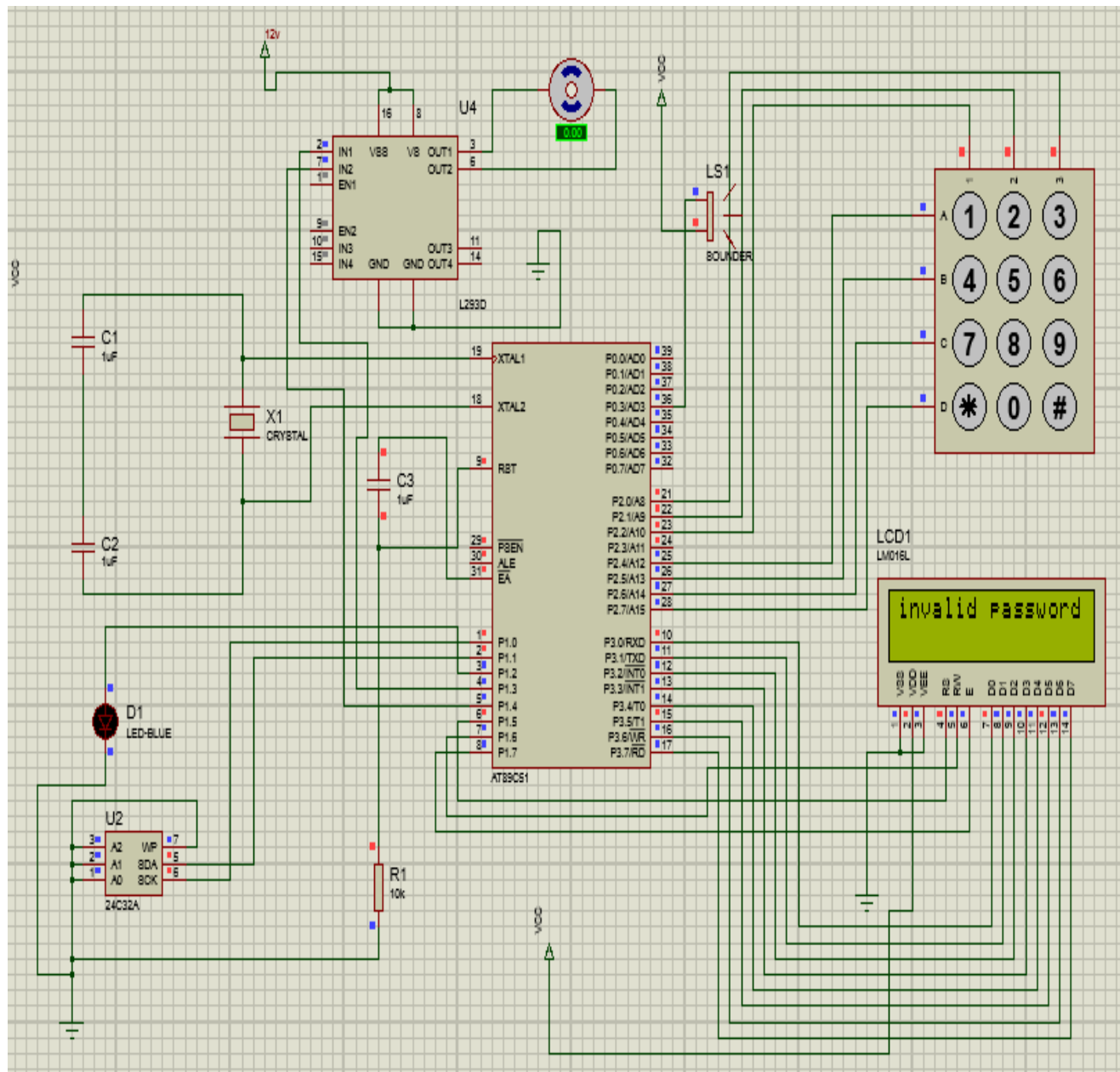


Figure 4.5: simulation result for invalid password for window

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CHAPTER FIVE

Conclusion and Future Recommendation

5.1 Conclusion

This project is meant for security systems whose access is only for respected authorities. Using a microcontroller the password entered is checked with the stored password and then does the corresponding operations. Here we use a 4 digit password for the door.

When we completed this project, we had learned and understand the process of constructing the home security system which contained hardware and software development. We also know and understand the microcontroller 8051 and how to implement it on our project. Designing code lock and IR sensor based security system is very interesting for us where we can get involve in programming environment seriously. Our digital code lock performed as expected.

We were able to implement all the functions specified in our proposal. The biggest hurdle we had to overcome with this project was interfacing the microcontroller with the hardware components. We feel that this digital code lock and infrared based security control system is very marketable because it is easy to use, comparatively inexpensive due to low power consumption and highly reliable. This digital code lock and IR sensor based security control is therefore particularly useful in applications such as hotel room door locks, residential housing and even office buildings.

5.2 Future Recommendation

In meeting the design requirements, there are several technical challenges that must be overcome. In our project the challenging task was writing the program in keil uvision5. In this project we face different problems and constraints in which some of them have been met but there are different challenges that have been an obstacle to accomplish the software and hardware of our project part.

The big challenge was scarcity of materials for implementing our design in our laboratories. As we tried to state on our proposal this project needs some materials such as microcontroller, motor drive, sounder, keypad, LCD, EEPROM, LED, crystal oscillator, dc motor, capacitor and resistor etc. but it was impossible to find most of these equipment in our lab . This problem was an obstacle to do hard ware implementation.

Generally we recommend to our department in order to solve the above problem for the next generation. In addition to this we recommend:

1. If anyone have chance to design this project again, he/she will add a phone dialer chip and GSM modem. The system can be easily connected to the personal computer for further control.
2. We can use this system as an attendance register for the students to enter a class room with their respective password.
3. The keypad and LCD also offer great interface and users can be familiar with our system in less than few seconds.

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Appendix

C source code

```
#include <AT89X51.H>
#define EE_SCL P1_0
#define EE_SDA P1_1
#define LCD_E P1_7
#define LCD_RW P1_6
#define LCD_RS P1_5
#define LCD_DATA P3
#define buzz P0_3
#define m1 P1_3
#define m2 P1_4
#define led P1_2
#define keypad p2
char uid[]="1432";
void cmnwrt(int);
void datawrt(char);
char keypad_refresh();
void init();
void delay(unsigned i);
void get_sntnc(char *);
void print_msg(char*,char);
void EE_write(unsigned int, char);
char EE_read(unsigned int);
void EE_strt();
void EE_stop();
char EE_shin();
void sounder();
void EE_shout(char);
//-----
//-----
int main(){
char tmp[17],i,j;
init();
cmnwrt(0x80);
m1=0;
m2=0;
buzz=0;
led=0;
```

```
while(1){
if(EE_read(0) != 'p' || EE_read(1) != '='){
np:
cmnwrt(1);
print_msg("Enter new password:\0",0x80);
get_sntnc(tmp);
EE_write(0,'p');
EE_write(1,'=');
for(i=0; tmp[i]; i++)
EE_write(i+2,tmp[i]);
EE_write(i+2,0);
}
rpa:
cmnwrt(1);
print_msg("Enter password:\0",0x80);
get_sntnc(tmp);
for(i=0,j=0;;i++){
if(tmp[i] != EE_read(i+2)){
j=-1;
break;
}
if(!tmp[i])
break;
}
cmnwrt(1);
if(j== -1)
{
print_msg("invalid password!\0",0x80);
sunder();
m1=0;
m2=0;
led=0;
delay(500);
buzz=1;
delay(1000);
buzz=0;
goto rpa;
}
else
{
```



```
print_msg("valid Password!\0",0x80);
delay(1000);
print_msg("door opening...\0",0x80);
m1=1;
m2=0;
led=1;
buzz=0;
delay(10000);
m1=0;
m2=1;
led=1;
delay(500);
print_msg("door closing...\0",0x80);
delay(10000);
m1=0;
m2=0;
led=1;
delay(1000);
}
cmnwrt(1);
print_msg("*=change password\0",0x80);
print_msg("#=back\0",0xc0);
ka:
do
i=keypad_refresh();
while(i!=-1);
if(i==0x3a)
goto np;
else if(i==0x3c)
goto rpa;
else
goto ka;
}
}
//-----
//-----
void print_msg(char *msg, char line){
char i;
cmnwrt(line);
for(i=0; *(msg+i); i++)
```

```
datawrt(*(msg+i));
}
//-----
//-----
char keypad_refresh(){
char tmp,row,cl;
P2 = 0xf0;
if(!P2_4)
row = 0;
else if(!P2_5)
row = 1;
else if(!P2_6)
row = 2;
else if(!P2_7)
row = 3;
else
return -1;
P2 = 0xff;
P2 = 0x0f;
if(!P2_2)
cl = 0;
else if(!P2_1)
cl = 1;
else if(!P2_0)
cl = 2;
else
return -1;
tmp = row*3+cl+1;
if(tmp == 11)
tmp = 0;
return tmp|0x30;
}
//-----
//-----
void get_sntnc(char *str) {
char i,j,z;
for(i=0; i<17; i++)
*(str+i) = 0;
i=0;
while(1){
```

```
cmnwrt(0xc0);
for(z=0; z<16; z++)
if(*(str+z))
datawrt(*(str+z));
else
datawrt(' ');
delay(70);
do
j = keypad_refresh();
while(j==1);
if(j==0x3c){
*(str+i) = 0;
break;
}else if(j==0x3a){
if(i>0){
i--;
*(str+i) = 0;
}
}else{
if(i<=15){
*(str+i) = j;
i++;
}
}
}
}
void sounder()
{
int i;
for(i=0;i<10;i++)
{
buzz=1;
delay(100);
buzz=0;
}
}
//-----
//-----
void init() {
int i;
```

```
P0=P1=P2=P3=0;
//initialize LCD
LCD_RS = 0;
LCD_RW = 0;
delay(30);
cmnwrt(0x3f);
delay(20);
cmnwrt(0x3f);
delay(10);
cmnwrt(0x3f);
delay(5);
cmnwrt(0x38);
cmnwrt(0x01);
cmnwrt(0x0c);
//initialize EEPROM
EE_SDA = 1;
for (i=0; i<10; i++) {
EE_SCL = 1;
i=i; //delay
EE_SCL = 0;
}
}
//-----
//-----
void delay(unsigned i) {
    TMOD = 1;
    for (;i>0;i--) {
        TH0 = 0xFC; //0xFFFF - 0xFC17 = 1000 (decimal) -> 1ms for 12MH crystal
        TL0 = 0x17;
        TR0 = 1;
        while(!TF0);
        TR0 = 0;
        TF0 = 0;
    }
}
//-----
//-----
void cmnwrt(int cmn) {
    LCD_DATA = cmn;
    LCD_RS = 0;
```

```
LCD_RW = 0;
LCD_E = 1;
LCD_E = 0;
delay(2);
}
void datawrt(char ch) {
    LCD_DATA = ch;
    LCD_RS = 1;
    LCD_RW = 0;
    LCD_E = 1;
    LCD_E = 0;
    delay(2);
}
//-----
//-----
//Write into EEPROM
void EE_write(unsigned int addr, char dta){
    char chtmp;
    CY = 0;
    EE_strt();
    EE_shout(0xA0);
    chtmp = addr>>8;
    EE_shout(chtmp);
    chtmp = addr;
    EE_shout(chtmp);
    EE_shout(dta);
    EE_stop();
    delay(13);
}
//Reading from EEPROM
char EE_read (unsigned int addr) {
    char chtmp;
    CY = 0;
    EE_strt();
    EE_shout(0xA0);
    chtmp = addr>>8;
    EE_shout(chtmp);
    chtmp = addr;
    EE_shout(chtmp);
    EE_strt();
```

```
EE_shout(0xA1);
  chtmp = EE_shin();
EE_stop();
return chtmp;
}
//Start
void EE_strt() {
EE_SDA = 1;
EE_SCL = 1;
EE_SDA = 0;
EE_SCL = 0;
}
//Stop
void EE_stop() {
EE_SCL = 0;
EE_SDA = 0;
EE_SCL = 1;
EE_SDA = 1;
}
//Shift out
void EE_shout(char sho) {
  unsigned char i,j;
  for (j=0, i=0; i<8; i++) {
j /= 2;
if (!j)
j=128;
if (sho&j)
CY = 1;
else
CY = 0;
EE_SDA = CY;
EE_SCL = 1;
j = j; //delay
EE_SCL = 0;
}
EE_SDA = 1;
EE_SCL = 1;
j=j; //delay
CY = EE_SDA;
EE_SCL = 0;
```

```
}  
//Shift in  
char EE_shin() {  
    char i,shi = 0;  
    EE_SDA = 1;  
    for (i=0; i<8; i++) {  
        CY = 0;  
        shi <<= 1;  
        EE_SCL = 1;  
        i=i;  
        CY = EE_SDA;  
        EE_SCL = 0;  
        if (CY)  
            shi++;  
    }  
    EE_SDA = 1;  
    EE_SCL = 1;  
    i=i; //delay  
    EE_SCL = 0;  
    return shi;  
}
```